Cloud Forensics: Tracing Digital Footprints in the Cloud

Tools, Techniques, and Tactics for Forensic Analysis in the Cloud

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JØRN CONRADI

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1. Introduction

1.1 Introduction to Cloud Forensics

Cloud forensics is a part of digital forensics, and it involves using investigation methods to collect, study and understand the digital evidence that is kept in the cloud. The significance of cloud forensics has increased because of the rising popularity of cloud services. This is because cloud forensics guarantees that digital evidence is managed appropriately for legal cases and responding to security incident responses.

(Oxygenforensics.com, 2024)

In today's digital world, a lot of data is stored and handled in the cloud. Cloud forensics is extremely important, as it assists organizations in dealing with security incidents, data breaches, and fulfilling legal obligations.

Being able to examine and understand data in cloud environments is crucial for maintaining the integrity and security of information systems.

(Oxygenforensics.com, 2024)

1.2 Objective of the Project

This project is based on a fictive situation, and it simulates a malicious insider from the company “Conradi Consulting”.

A company called Conradi Consulting suspect that their resources are being compromised. They assume there is a malicious insider in the company, but there is no evidence to back this hypothesis. The company’s desire is to find out who the perpetrator is with the tools and techniques of cloud forensics.

The objective of this project is to demonstrate the tools that can be used for cloud forensics using Microsoft Azure as the platform. The project will focus on connecting an Azure storage account to a Log Analytics workspace and use the Kusto Query Language (KQL) to query Azure storage logs. It will also concentrate on how to collect a full disk image of an Azure Virtual Machine, (VM) and investigate this image in Autopsy.

Finally, a comprehensive forensic report will be written based on the digital evidence acquired throughout the investigation process. This report will provide a detailed analysis of the findings.

1.3 Assumptions

It is assumed that the reader of this project is familiar with cloud computing and forensic concepts. It is also assumed that they have access to an Azure subscription with administrative privileges for following along with the practical tasks.

1.4 Expected Challenges

There are some potential challenges that readers may encounter while following the practical task, such as navigating the Azure portal interface, understanding the terminology, or troubleshooting technical issues. Readers who are new to forensic analysis might have a challenge of interpreting and analyzing log data.

1.5 Overview of Chapters

- Chapter 2 explains what insider threats are and what security risk these threats pose for companies. The chapter concludes by discussing how organizations can mitigate these risks.
- Chapter 3 provides case studies of real-life insider threats.
- Chapter 4.1 to 4.4 is where the practical tasks are solved.
- Chapter 5 concludes with a forensic report of the simulated attack.
- Chapter 6 contains a discussion and analysis of the findings and lessons learned.
- Chapter 7 reflects on the findings and reaches a conclusion.
- Chapter 8 lists all the sources gathered for this project.

1.6 How the Practical Part was Solved

The simulated threat scenario by the malicious insider was completed by setting up an environment in Azure. A new user was created in Microsoft Entra ID for the insider to carry out the malicious activity. A storage account was created, and a document containing "confidential" information was uploaded to the storage account.
Finally, a virtual machine was created for the insider to log into.

Chapter 5 of the book *Cloud Forensics Demystified* (Ramakrishnan and Haqanee, 2024) was used to complete Chapter 4.1 and 4.2 of this project.

Chapter 10 of the same book has been used to complete Chapter 4.3 within this project.

The official Autopsy user documentation was utilized for Chapter 4.4. (Sleuthkit.org, 2016)

2. Examining Insider Threats

2.1 What are Insider Threats?

Insider threats are defined by the Cybersecurity and Infrastructure Security Agency (CISA) as:

> *"The threat that an insider will use their authorized access, intentionally or unintentionally, to do harm to the department's mission, resources, personnel, facilities, information, equipment, networks, or systems."* (Cisa.gov, 2021)

An insider threat refers to a security risk that comes from within the organization itself. It involves individuals who are allowed access to the organization's resources and misuse this access for harmful purposes.

These actions can be deliberate or accidental and might involve behaviors such as stealing, spying, sabotaging, or compromising information and systems.

Insider threats can come from current or former employees, contractors, or business partners with internal knowledge of the organization's security policies, data, and computer systems.

The damage caused by insider threats can impact the integrity, confidentiality, and availability of an organization's data, staff, or premises. (Cisa.gov, 2021)

2.2 What are the Different Types of Insider Threats?

Some examples of insider threats include *unintentional threats* and *intentional threats.*

An unintentional threat can be negligence. This type of insider poses a threat due to carelessness. Negligent insiders are aware of security policies but neglect them, putting the organization at danger. Some examples of this can be allowing unauthorized personnel to enter secure locations, which is known as "piggybacking", ignoring

requests to update security software, or losing a storage device like USB with sensitive data.

Another unintentional threat is accidental. This person unintentionally creates risks for the organization by sending sensitive documents to the wrong recipient or clicking on malicious URLs from a phishing email. (Cisa.gov, 2021)

Intentional threats are commonly known as a malicious insider. These intentional threats from the insider are aimed at damaging a company for personal gain or in response to personal resentment.

Like for instance, some insiders seek revenge after feeling unappreciated or being fired. They may commit sabotage, stealing intellectual property, or leaking sensitive data to promote their own careers. (Cisa.gov, 2021)

A 2023 Report by Cybersecurity Insiders revealed that nearly 74% of organizations face some level of vulnerability to insider threats. (Webb, 2023)

2.3 How to Mitigate Insider Threats

Organizations can mitigate insider threat risks by using a combination of techniques that focus on identifying, preventing, and responding to these threats. The following are five best practices that companies can implement to mitigate insider threats:

1. The majority of insiders fall under the "careless" category. It is critical to identify users who might pose a threat due to their access to sensitive data or their behavior.
2. Encouraging effective communication regarding security policies helps mitigate the risk of unintentional insider threats.
3. Visibility is crucial. By understanding the context in which users operate can help companies to successful threat prevention and mitigation.
4. Educating employees on how to recognize and avoid risky behaviors through a security awareness program can help prevent security issues.
5. Establishing a proactive response plan for insider threats can mitigate damage and promote quick recovery. (Torto, 2023)

3. Case Studies

This section will delve into real-life cases of insider threats, examining the specific incidents, and the methods they used to exploit their access.

3.1 The Data Breach of Cash App's Customers

Cash app is a mobile payment service that experienced a data breach by a former disgruntled employee. In the month of April 2022, the employee downloaded personal data of the customers of Cash App including full names, brokerage portfolio values, holdings, and stock trading activity.

The breach resulted in 8.2 million users being affected. Because impacted consumers weren't informed until four months after the breach, Cash App Investing and its parent company, Block, were sued in a class action lawsuit.

The employee was terminated on December 10, 2022, but retained access due to the company's failure to revoke permissions.

To avoid this situation the company could have developed an appropriate termination procedure for the employees, and also implement continuous activity monitoring and regular access reviews. (Pryimenko, 2024)

3.2 Former Yahoo Employee Accused of Stealing Trade Secrets

In February of 2022, Yahoo accused their former employee Qian Sang, of stealing their intellectual property. Yahoo claimed that Qian Sang intended to use the stolen information to benefit financially from Yahoo's rival, The Trade Desk. Before this incident happened, Sang received a job offer from The Trade Desk.

According to Yahoo, Sang also stole additional confidential information such as Yahoo's strategic plans and a competitive study of their rival, The Trade Desk.

Yahoo conducted a forensic investigation and found that Sang had downloaded 570,000 files including sensitive information and the source code of Yahoo's AdLearn engine. Yahoo claims the stolen data could give The Trade Desk a competitive edge in online advertising, and that this could potentially cause financial losses for Yahoo.

The forensic investigation by Yahoo revealed that Sang transferred the data from his work laptop to personal external storage devices. Yahoo sued their former employee Sang, arguing that the theft could have been prevented with better security tools such as employee monitoring software, USB device management, and real-time user activity alerts. These tools could have detected suspicious activities and communications, allowing the security team of Yahoo to respond before any data was stolen. (Pryimenko, 2024)

3.3 The Trade Secret Theft of Waymo

Waymo, a firm established by Google, develops autonomous vehicles, or self-driving cars. Waymo's lead engineer Anthony Levandowski left the company in 2016 and he founded the company Otto, which also produces self-driving cars.

The transportation company Uber bought up Levandowski's company Otto, and in doing so they obtained the trade secrets that Levandowski had stolen from Google. Some of these secrets includes confidential documents, source code snippets, the marketing information of Waymo, and diagrams related to Light Identification Detection and Ranging (LIDAR) technologies.

According to an investigation, Levandowski was not satisfied while he worked at Google, and his actions at Google were intentional. Levandowski began to talk about leaving Google in 2015 and started to recruit his colleagues by trying to get them to join his new company. The truth was uncovered to Google executives when Uber bought up Levandowski's new firm Otto.

Levandowski downloaded 14,000 files from Google's servers onto an external hard drive shortly before resigning. This was Google's intellectual property stolen by Levandowski.

Waymo successfully proved the intellectual property theft, and as a compensation they were given Uber shares worth 245 million dollar. Uber committed themselves to not use the stolen secrets for their hardware or software development. (Exabeam.com, 2022)

3.4 The Negligent Employees at Microsoft

Some insider threats do not have a malicious intent, like what happened at Microsoft in August of 2022.

Lots of Microsoft employees unintentionally leaked the login credentials to their own GitHub infrastructure. If these credentials were exploited, they could have granted unauthorized access to Azure servers and potentially other critical internal systems within Microsoft.

The exposure of such sensitive information, including Microsoft source code, posed significant risks to both the company and its customers. Microsoft did not specify which systems the compromised credentials protected, but there was a potential for an outsider to exploit this vulnerability.

If the customer information from the European Union (EU) had been exposed, Microsoft could have faced severe repercussions, including heavy fines under the GDPR regulations.

The good news for Microsoft was that the exposed credentials was detected by the cybersecurity company spiderSilk, and they quickly informed Microsoft of the security breach.

Microsoft is now taking proactive measures to enhance security protocols and prevent similar incidents from recurring in the future. (Ostendorf, 2023)

3.5 Boeing Employee Convicted of Theft of Trade Secrets

A former engineer from Orange County, California, named Dongfan Chung, was found guilty of economic espionage and acting as an agent of China. He worked for Rockwell and Boeing, where he stole restricted technology and trade secrets, such as information about the Space Shuttle program and Delta IV rocket.

Chung's actions compromised national security and he was caught in 2006 after a three-year investigation by the FBI, NASA, and Boeing. He had exchanged letters with Chinese officials, made trips to China, and stored thousands of documents from Boeing and Rockwell at his home. Based on the evidence that was put forward during the trial, the earliest letters Chung had received from China dated all the way back to 1979.

Chung faced severe penalties, potentially up to 15 years in prison for each charge of economic espionage. (Justice.gov, 2009)

4. Demonstrations of Investigative Tools

This chapter demonstrates how incident responders can take advantage of tools such as Log Analytics and Autopsy to investigate digital crime in the cloud.

Figure 4 1 - Log Analytics logo.

Figure 4 2 - Autopsy logo.

Significance of the Practical Tasks

By setting up monitoring of an Azure storage account and querying the logs, incident responders can get all operations within a storage account. They will be able to detect and investigate security incidents from the Azure storage logs.

Through acquiring a full disk image from a virtual machine and download the image from the Azure portal, incident responders are able to investigate it further using the forensic tool Autopsy.

4.1 Setting Up Monitoring of Azure Storage

Log Analytics in Azure is a feature within Azure Monitor that enables companies to edit and execute log queries on data collected by Azure Monitor Logs. (Wild, *et al.* 2023)

A Log Analytics workspace must be created first to enable the logs being sent to Log Analytics.

In the Azure portal, search for **Log Analytics** and select it. Click on **Create** to create a new Log Analytics workspace.

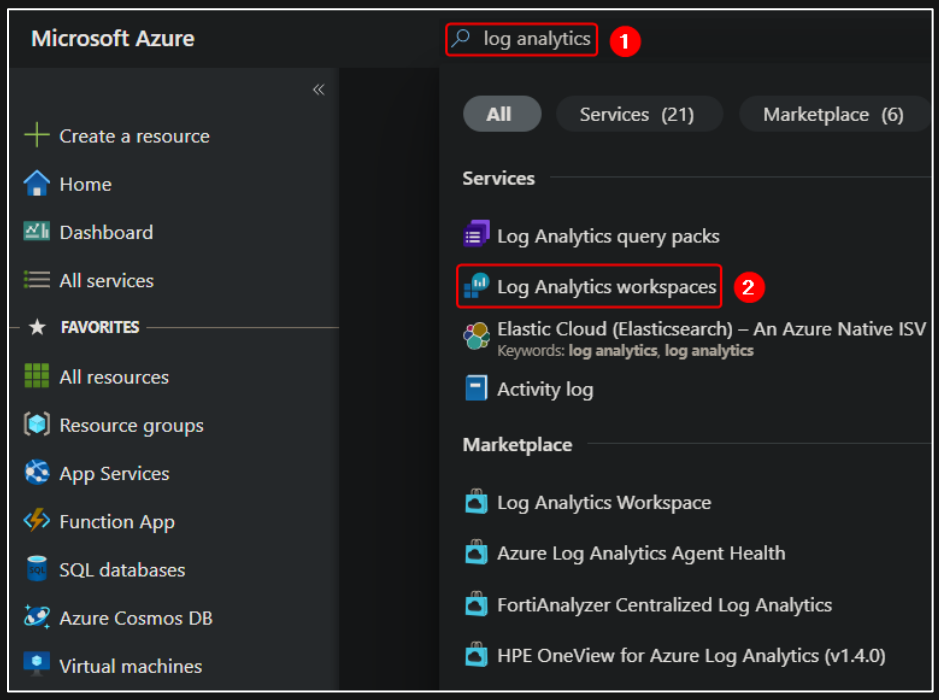


Figure 4.1 1 - Searching for Log Analytics.

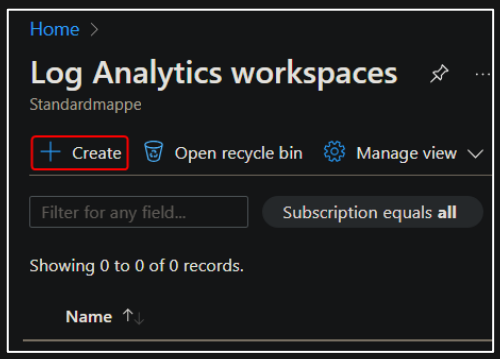


Figure 4.1 2 - Creating the workspace.

Choose a resource group and give a name for the Log Analytics workspace.
Select **Review + Create**, then **Create** to make the new workspace.

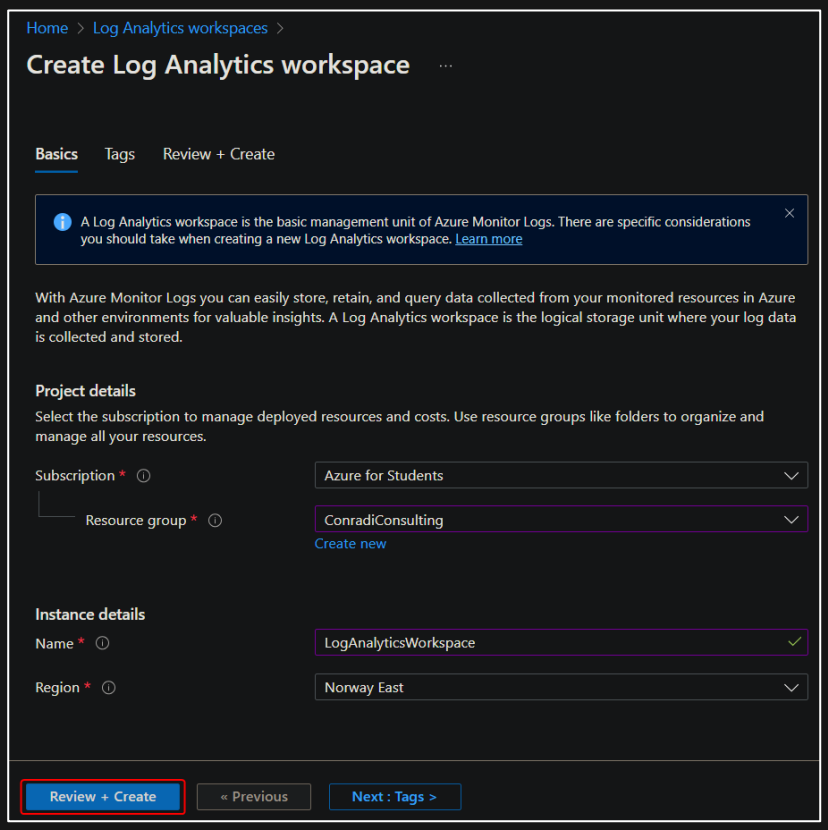


Figure 4.1 3 - Create new workspace.

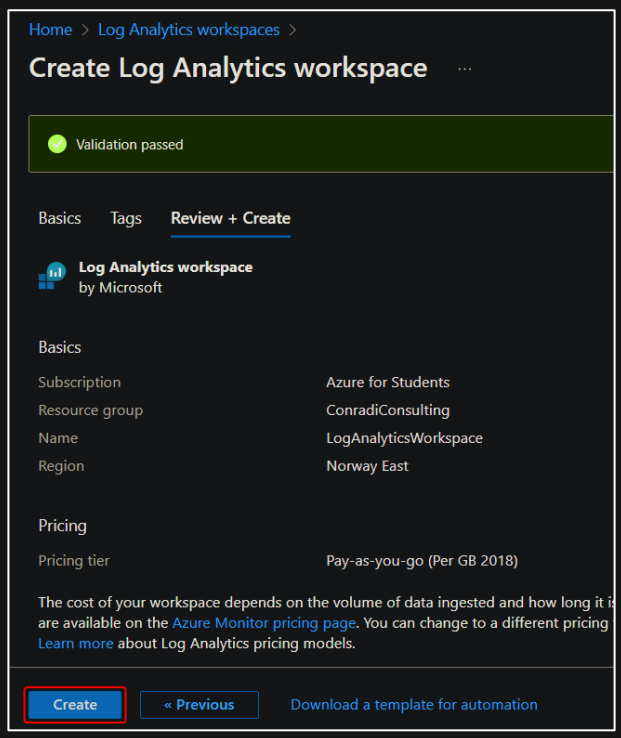


Figure 4.1 4 - Create new workspace (continued...)

Now that a Log Analytics workspace has been created, the storage account can be connected to Log Analytics.

First select **Storage accounts**, **Diagnostic settings** and then choose the name of the storage account (*figure 4.1.6*).

Figure 4.1 5 - Selecting the diagnostic settings.

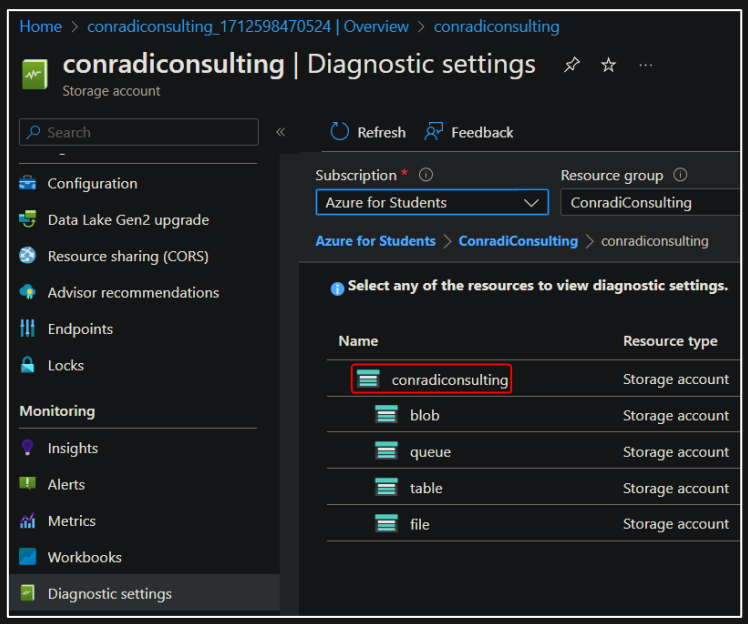


Figure 4.1 6 - Connecting storage resources to Log Analytics.

Select **Add diagnostic setting** to enable Log Analytics for the storage account.

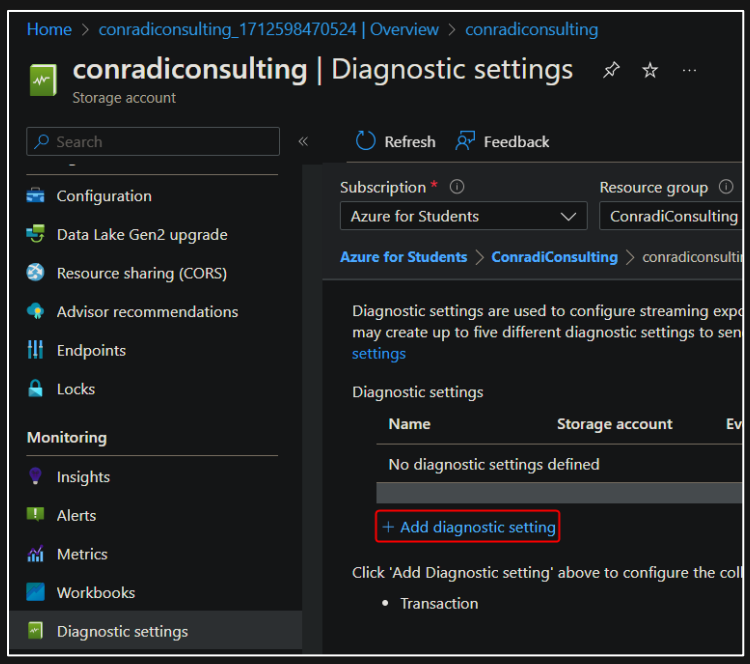


Figure 4.1 7 - Connecting storage resources to Log Analytics (continued...)

Figure 4.1.8 shows the diagnostic settings in detail, and the features and options that users can configure to meet their specific needs.

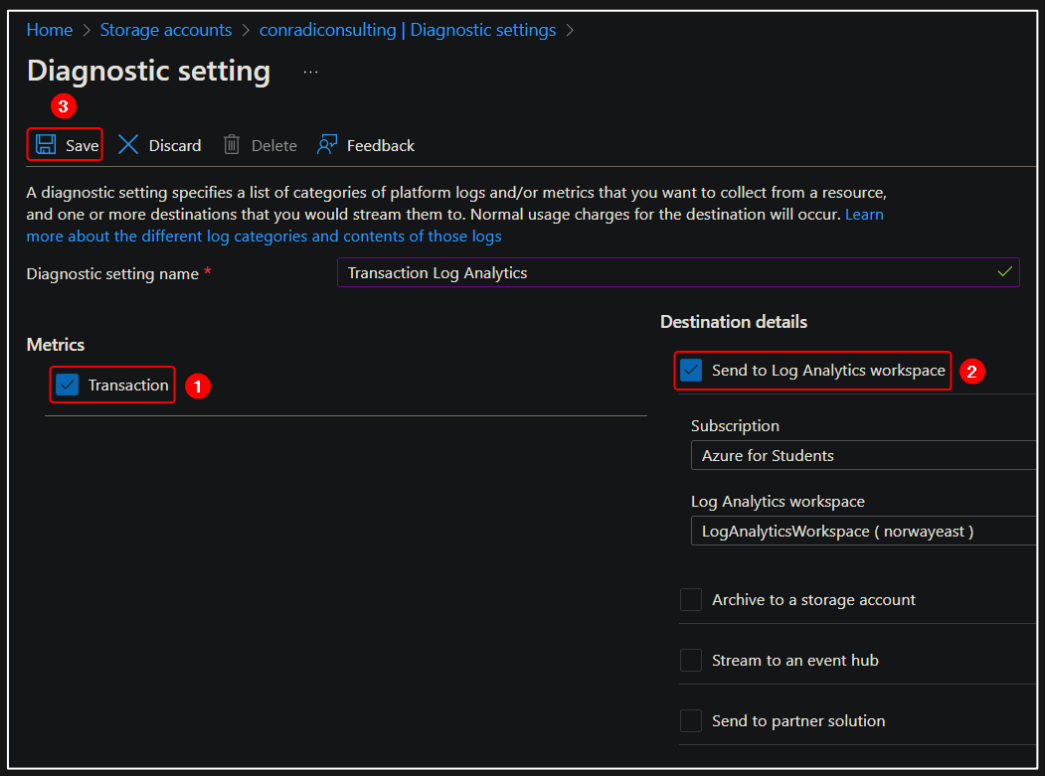


Figure 4.1 8 - Connecting storage resources to Log Analytics (continued...)

The same can be done to the Azure Blob level, so the read, write, delete and transaction logs gets directed to the Log Analytics workspace.

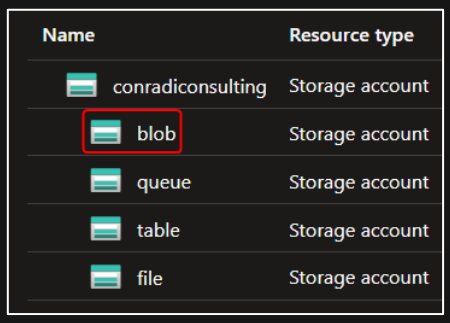


Figure 4.1 9 - Blob storage.

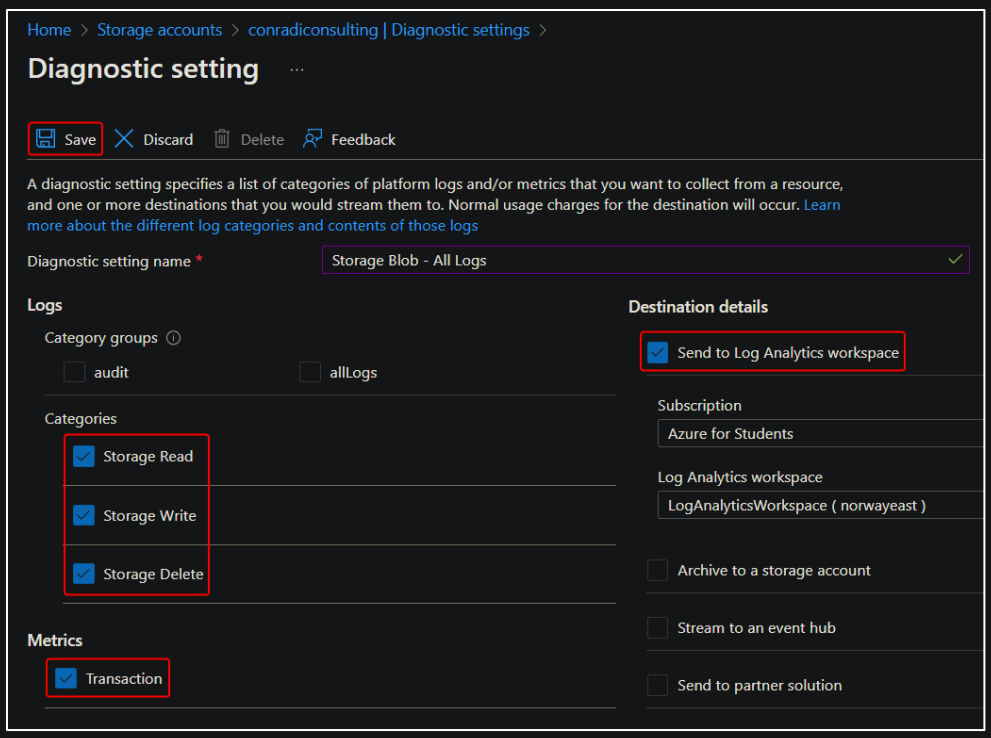
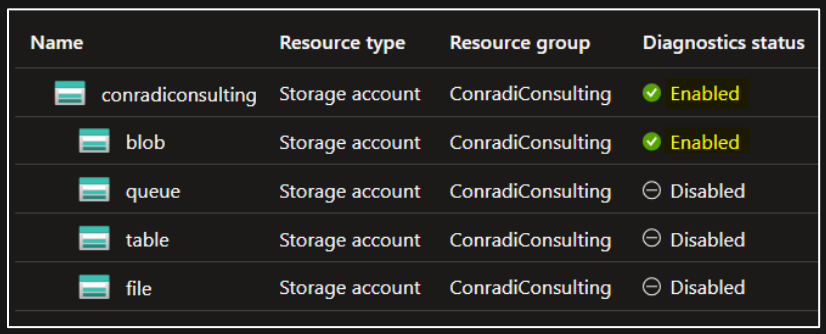


Figure 4.1 10 - Connecting storage resources to Log Analytics (continued...)

The logs for the resources have been successfully enabled, as seen in *figure 4.1.11.*



Name	Resource type	Resource group	Diagnostics status
conradiconsulting	Storage account	ConradiConsulting	Enabled
blob	Storage account	ConradiConsulting	Enabled
queue	Storage account	ConradiConsulting	Disabled
table	Storage account	ConradiConsulting	Disabled
file	Storage account	ConradiConsulting	Disabled

Figure 4.1 11 - Log Analytics successfully connected.

Azure Blob Storage is a cloud service from Microsoft that lets organizations store large amounts of unstructured data, such as text files, images, and videos which don't follow a specific format or structure. (Jewell, *et al.* 2023)

4.2 Querying Azure Storage Logs for Forensic Analysis

After successfully setting up and configuring all necessary resources for the storage account, incident responders are now ready to start querying the logs associated with the storage account. The first step is to navigate to the storage account and then click on **Logs** under monitoring.

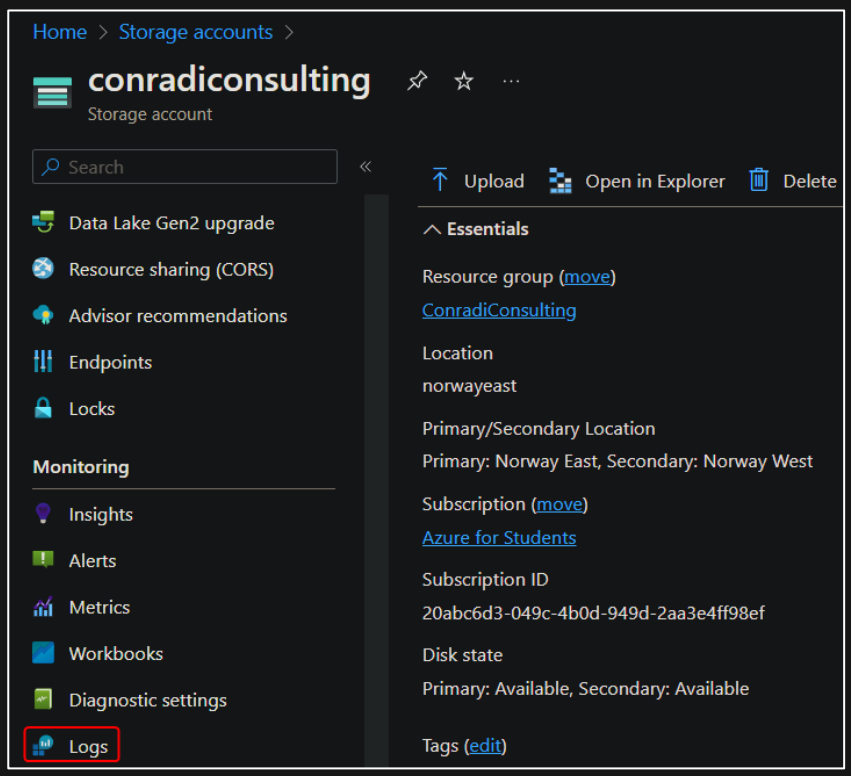


Figure 4.2 1 - Selecting the logs functionality.

Kusto Query Language

Azure uses the Kusto Query Language (KQL) to retrieve the data stored in Azure. With the help of KQL's syntax and set of operators, incident responders can analyze and retrieve this data. It is vital for incident responders to comprehend KQL to gain insight and get a visualization of all the data logs in Azure.
(Ramakrishnan and Haqanee 2024, p. 113)

Figure 4.2.2 displays the query panel where the logs for the storage account is stored.

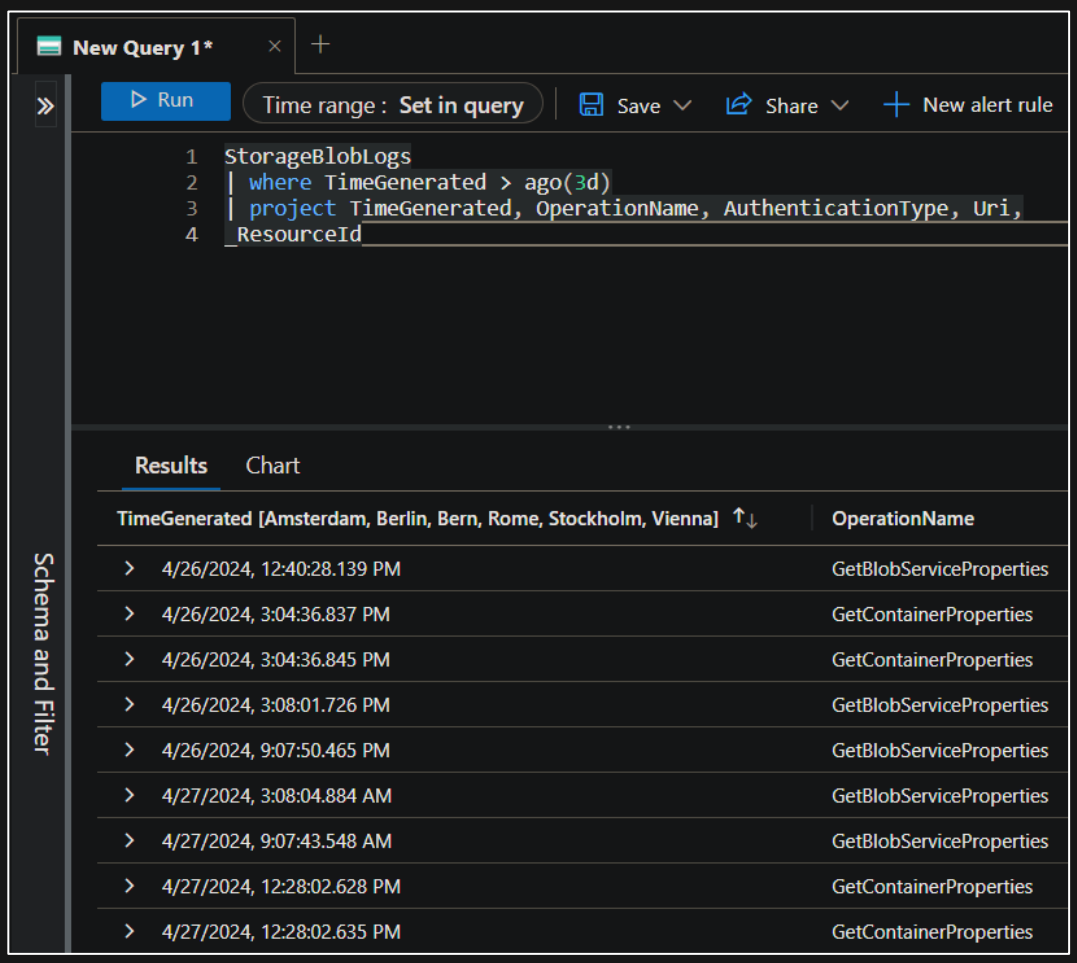


Figure 4.2 2 - The query panel.

When a user runs the query shown in *figure 4.2.3*, Azure displays all storage account operations seen in *figure 4.2.4.* (Ramakrishnan and Haqanee 2024, p. 133)

This allows investigators to find details such as file uploads, modifications, deletions, and access attempts. By taking advantage of Azure's query capabilities, incident responders can monitor storage activities in real-time, track changes made to data, and identify any unauthorized actions.

```
StorageBlobLogs
| where TimeGenerated > ago(7d)
| project TimeGenerated, OperationName, AuthenticationType, Uri, _ResourceId
```

Figure 4.2 3 - Storage logs query.

Results Chart

TimeGenerated [Amsterdam, Berlin, Bern, Rome, Stockholm, Vienna]	OperationName
4/26/2024, 12:40:28.139 PM	GetBlobServiceProperties
4/26/2024, 3:04:36.837 PM	GetContainerProperties
4/26/2024, 3:04:36.845 PM	GetContainerProperties
4/26/2024, 3:08:01.726 PM	GetBlobServiceProperties
4/26/2024, 9:07:50.465 PM	GetBlobServiceProperties
4/27/2024, 3:08:04.884 AM	GetBlobServiceProperties
4/27/2024, 9:07:43.548 AM	GetBlobServiceProperties
4/27/2024, 12:28:02.628 PM	GetContainerProperties
4/27/2024, 12:28:02.635 PM	GetContainerProperties

Figure 4.2 4 - Operations of the storage account.

Incident responders can use the query in *figure 4.2.5* to narrow down on the events and only show the operations that have failed. The `!contains` operator is used to filter out records that contain a specified substring.

When `!contains "Success"` is used, the query will return only the records where the field does not contain the substring "Success", in other words the failed operations. (Sagir, *et al.* 2024)

By implementing this, incident responders can find evidence of someone who may have breached a company's network or endpoint. This is known as an indicator of compromise (IOC). (Microsoft.com, 2023)

```
StorageBlobLogs
| where TimeGenerated > ago(3d) and StatusText !contains "Success"
| project TimeGenerated, OperationName, Uri, RequesterUpn
```

Figure 4.2 5 - Query that lists the failed operations.

By clicking the arrow highlighted in *figure 4.2.6*, the user gets a simple overview of the selected operation.

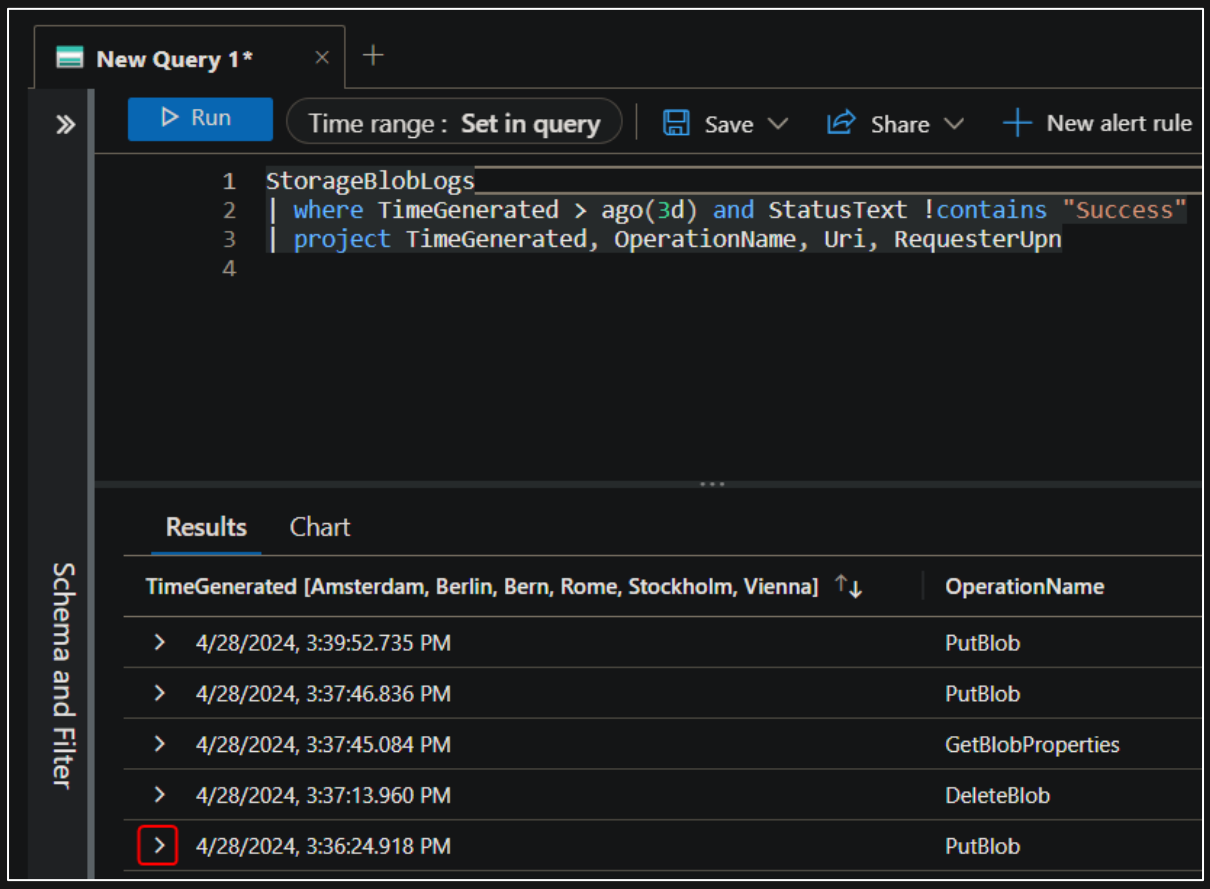


Figure 4.2 6 - Selecting the arrow button.

TimeGenerated [UTC]	2024-04-28T13:36:24.9181775Z
OperationName	PutBlob
Uri	https://conradiconsulting.blob.core.windows.net:443/documents/TradeSecrets.docx?_=1714311385185
RequesterUpn	Joe_Mole@jorn3006hotmail.onmicrosoft.com

Figure 4.2 7 - Details of a retrieved operation.

Chapter 5 will go more in depth into the evidence uncovered in the storage logs.

4.3 Forensic Acquisition of Virtual Machines

In Azure, users have the capability to capture a complete disk image of a virtual machine. By creating a snapshot of the virtual machine ensures that all data, configurations, and system settings are preserved intact. Once the snapshot is generated, users can easily download the disk image directly from the Azure portal.

To begin the acquisition of a compromised virtual machine, search for **snapshots** in the Azure portal.

Figure 4.3 1 - Searching for snapshots.

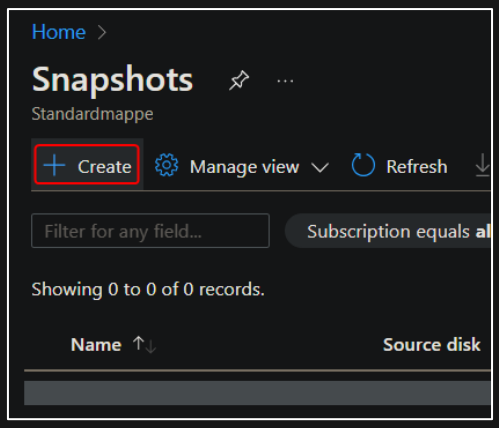


Figure 4.3 2 - Creating the snapshot.

When creating a snapshot of a virtual machine, make sure to choose the resource group where the VM resides (*figure 4.3.3*).

Select **Full** to capture the full disk image of the VM.

The **Source disk** is the VM disk for which the snapshot is created, so ensure that the correct disk is selected.

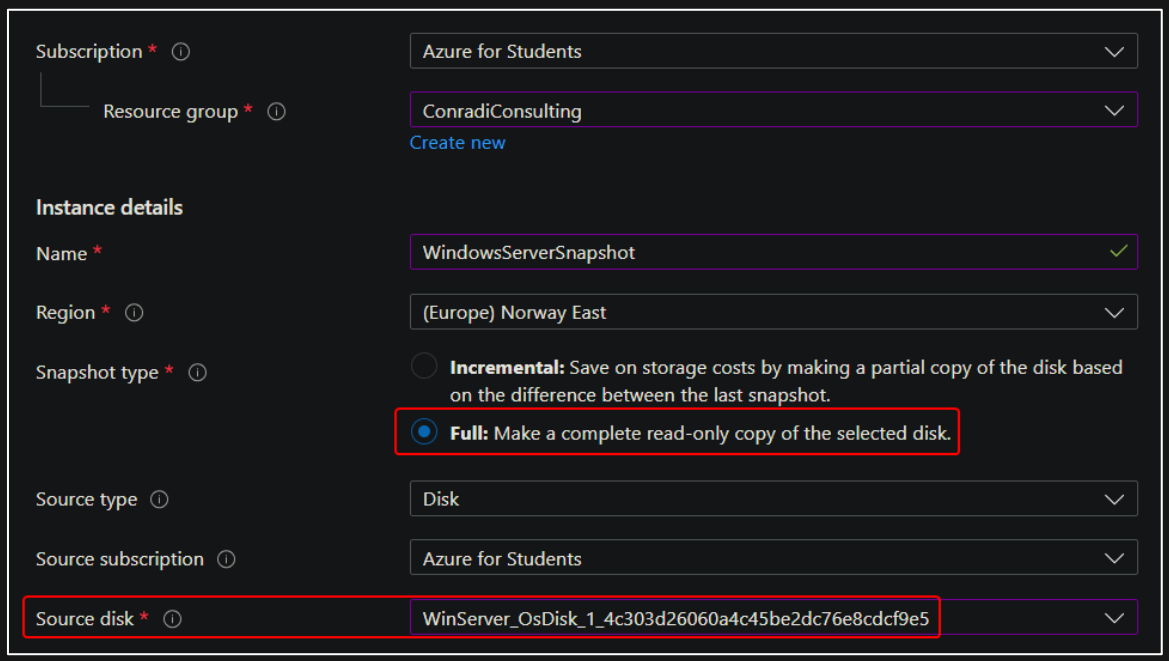


Figure 4.3 3 - Details of creating the virtual machine snapshot.

Click **Create** to make the snapshot of the disk and **Go to resource** when deployment is completed.

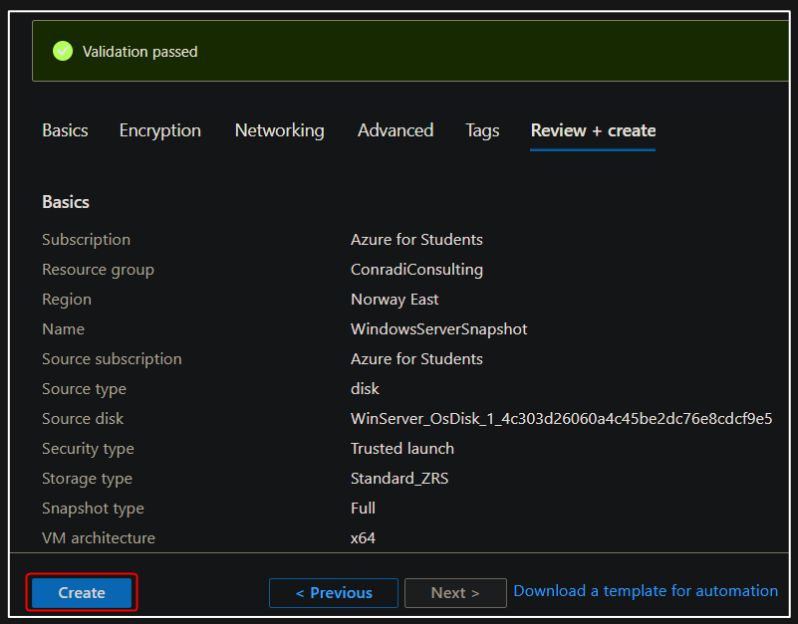


Figure 4.3 4 - Creating the snapshot (continued...)

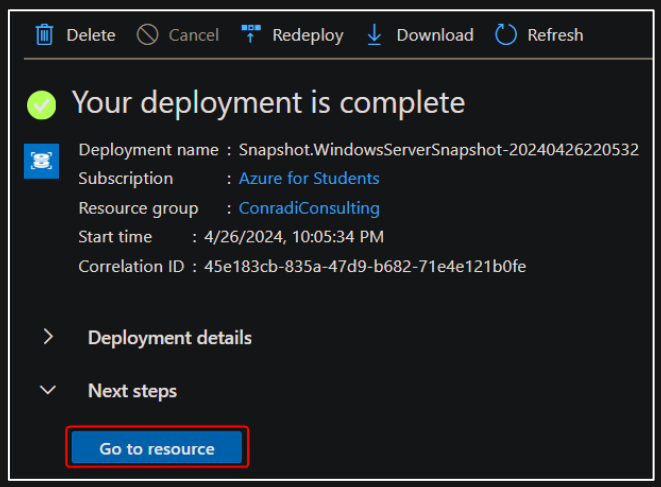


Figure 4.3 5 - Deployment of snapshot completed.

After the snapshot has been created, investigators can export and download the image directly from their browser.

In the left pane of the snapshot overview, navigate to **Settings** and select **Snapshot export.**

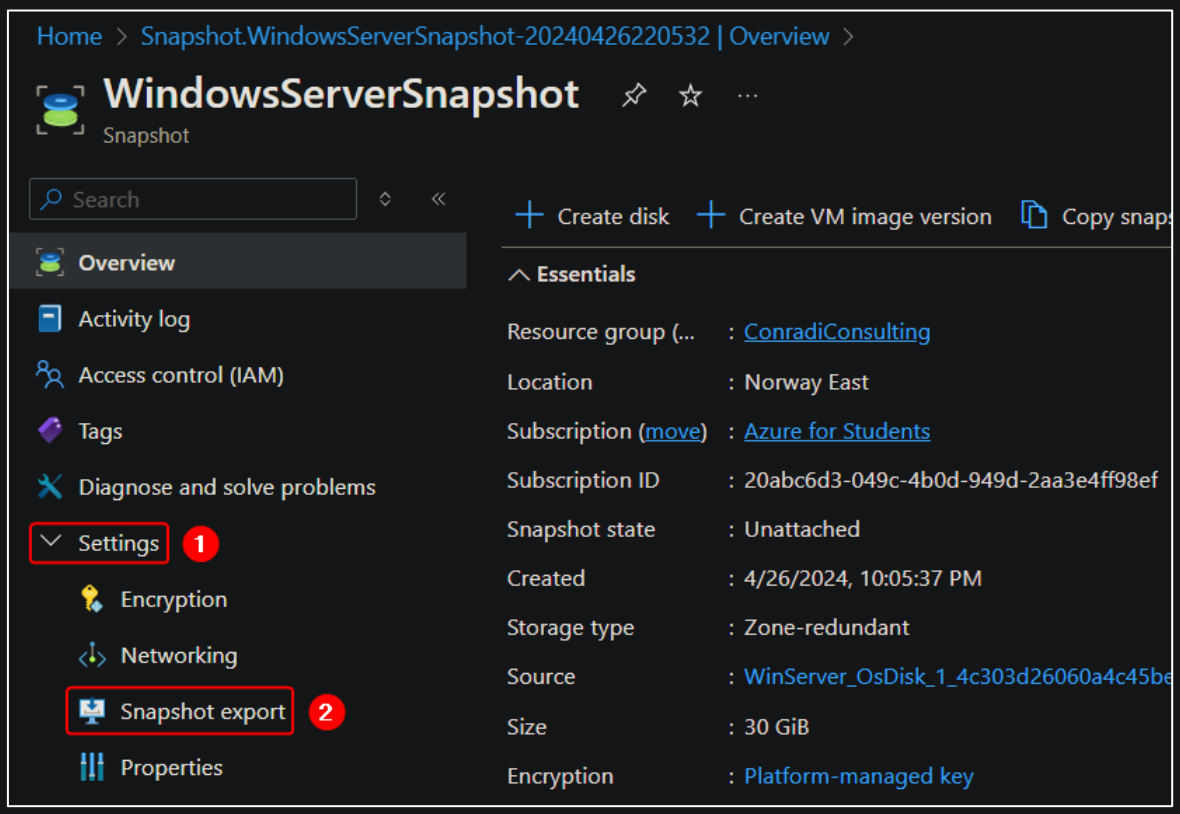


Figure 4.3 6 - Overview of the newly created snapshot.

Azure generates a secure URL that will expire, so investigators may need to choose the time it takes to export the snapshot. Azure allows a limit of 10 hours before the URL expires, which is 36,000 seconds.

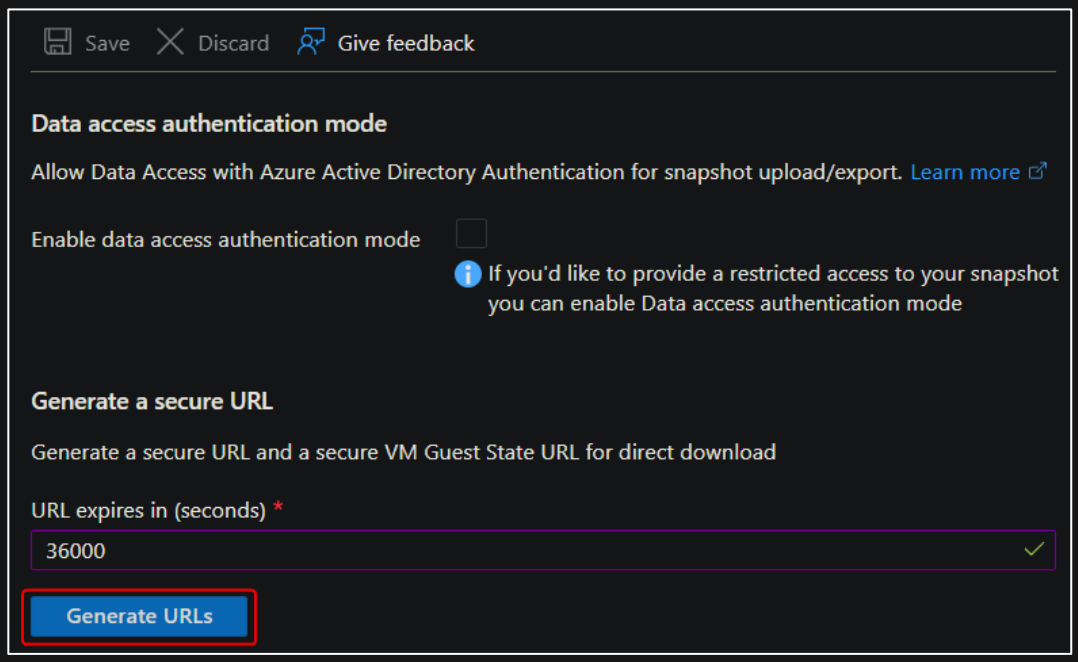


Figure 4.3 7 - Exporting the snapshot.

When selecting **Download the VHD file** in *figure 4.3.8*, the disk image starts downloading from the browser.

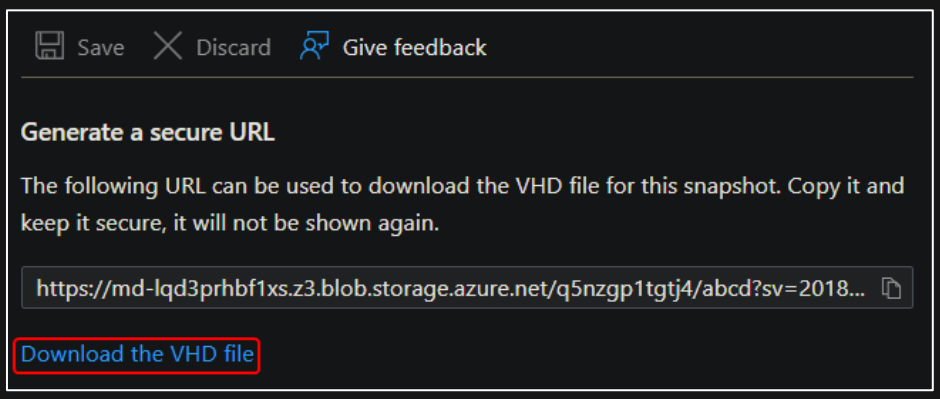


Figure 4.3 8 - Direct downloading of the VHD file.

4.4 Importing the Disk Image File in Autopsy

Autopsy is a digital forensic tool used by military and law enforcement to investigate compromised computers. By utilizing Autopsy, investigators can find out what has happened on a compromised computer. (Sleuthkit.org, 2006)

Before data can be analyzed in Autopsy a new case needs to be created first.
Select the **New Case** button.

The case information requires the user to enter a case name and a base directory for the new case.

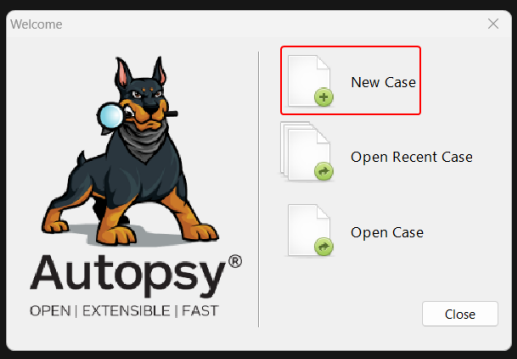


Figure 4.4 1 - Creating a new case.

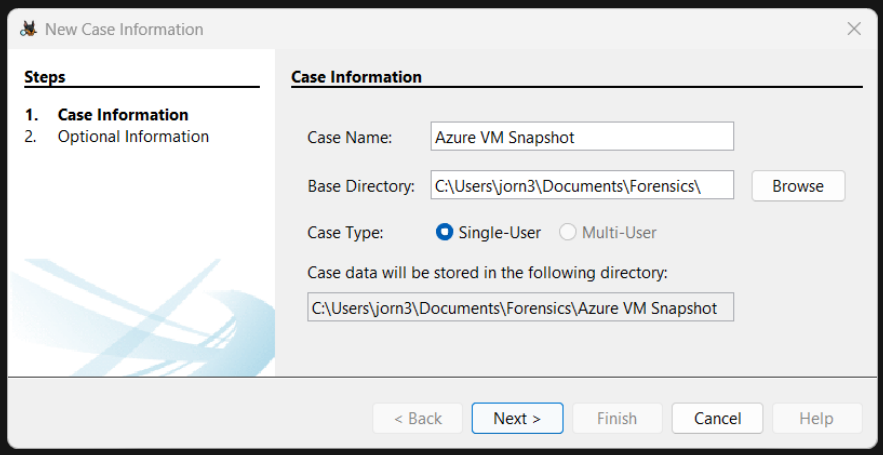


Figure 4.4 2 - Case information.

Choose a case number and information about the investigator.

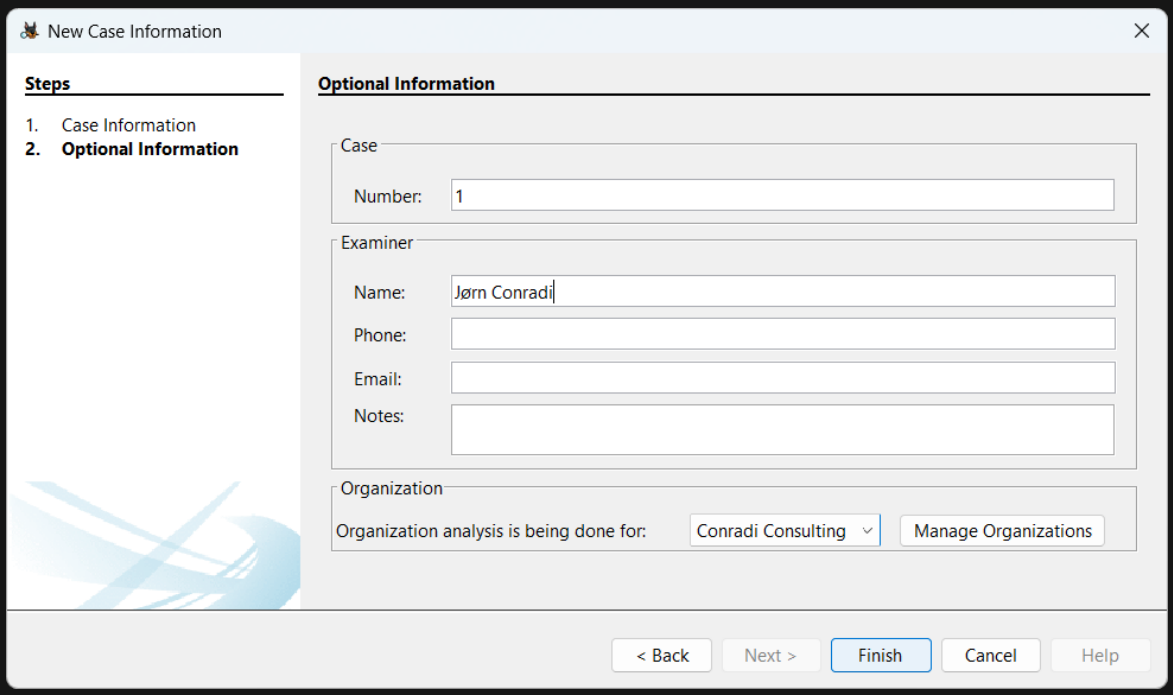


Figure 4.4 3 - Case information (continued...)

Next, select a new host name for the data source.

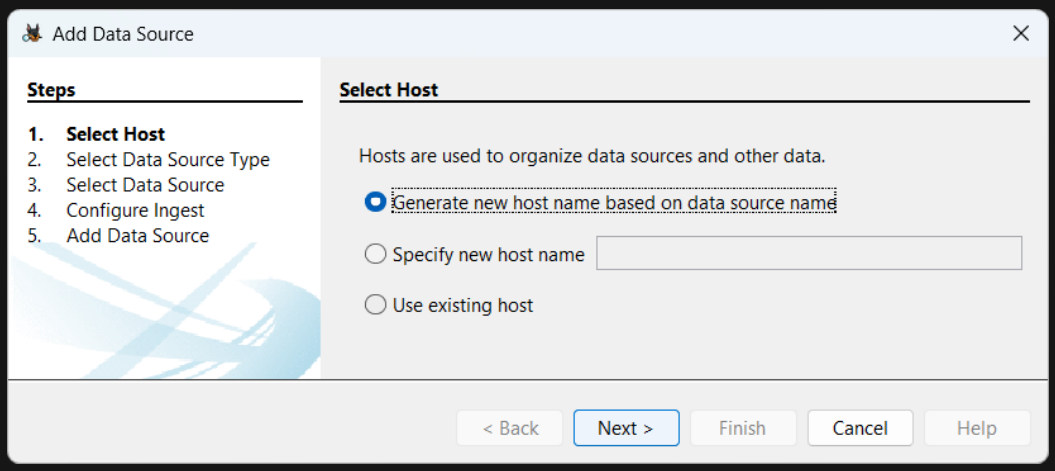


Figure 4.4 4 - Select the host name.

The user will be prompted to specify a data source, in this case it will be **Disk Image or VM File.**

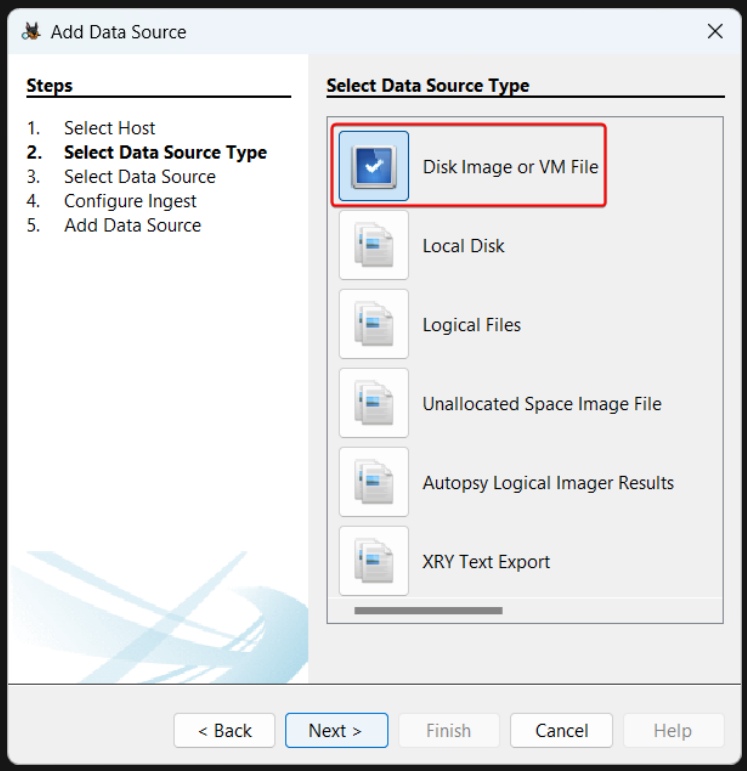


Figure 4.4 5 - Specifying the data source.

The next step is to locate the path of the disk image downloaded from Azure. Click the **Browse** button to select the location of the virtual hard disk (VHD) file.

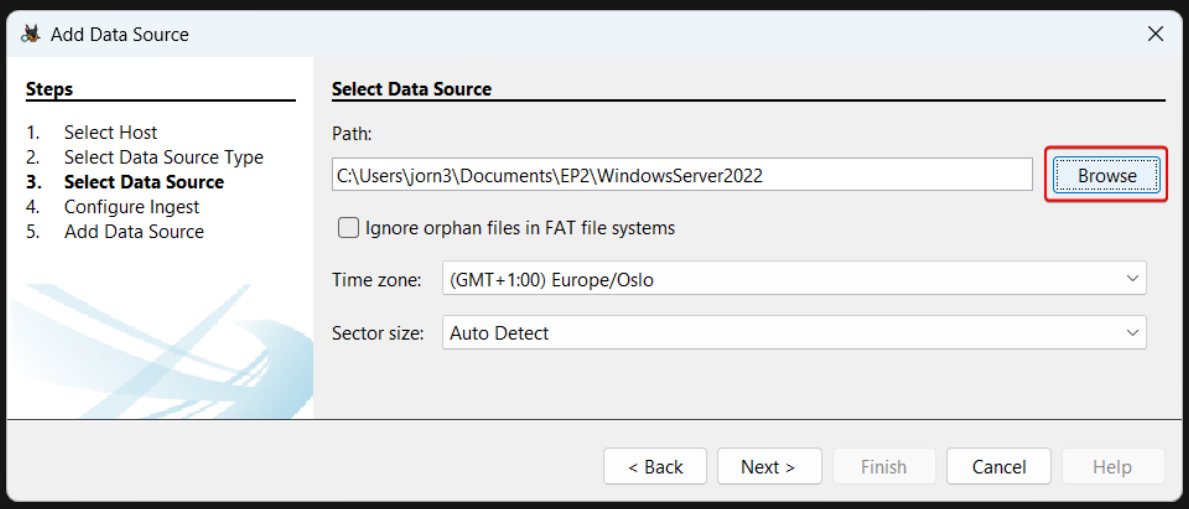


Figure 4.4 6 - Locating the data source.

The user can choose which ingest modules that Autopsy shall run.
For this project, the **Recent Activity** selection is sufficient.

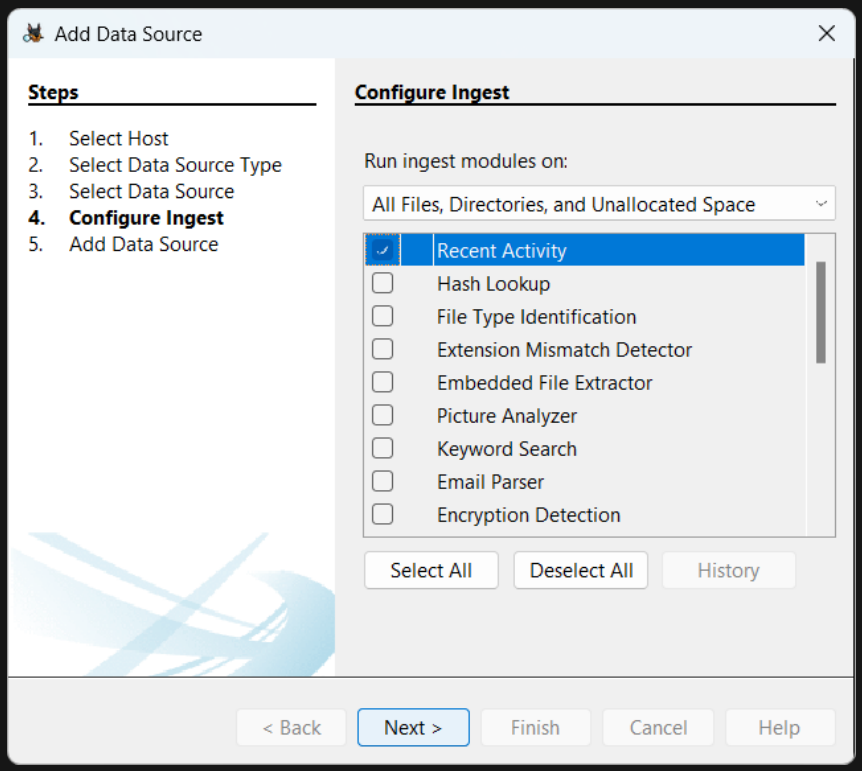


Figure 4.4 7 - Configuring the ingest modules.

Click the **Finish** button to complete the wizard. Autopsy will now examine the data source. This may take a time depending on the size of the disk.

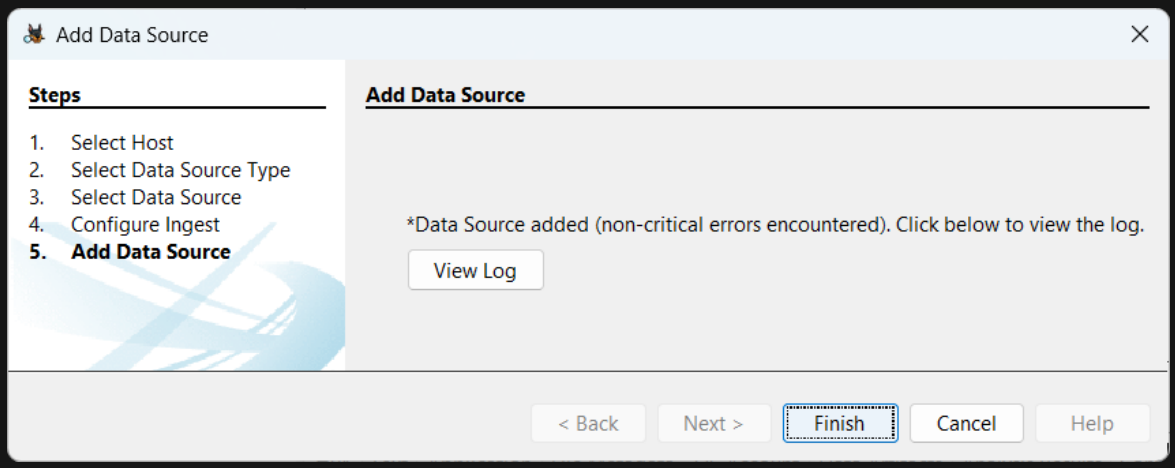


Figure 4.4 8 - Finished adding the data source.

Now that Autopsy is finished examining the data source, investigators can begin analyzing the information.

In the left pane of *figure 4.4.9* is where Autopsy has found results. These are called data artifacts.

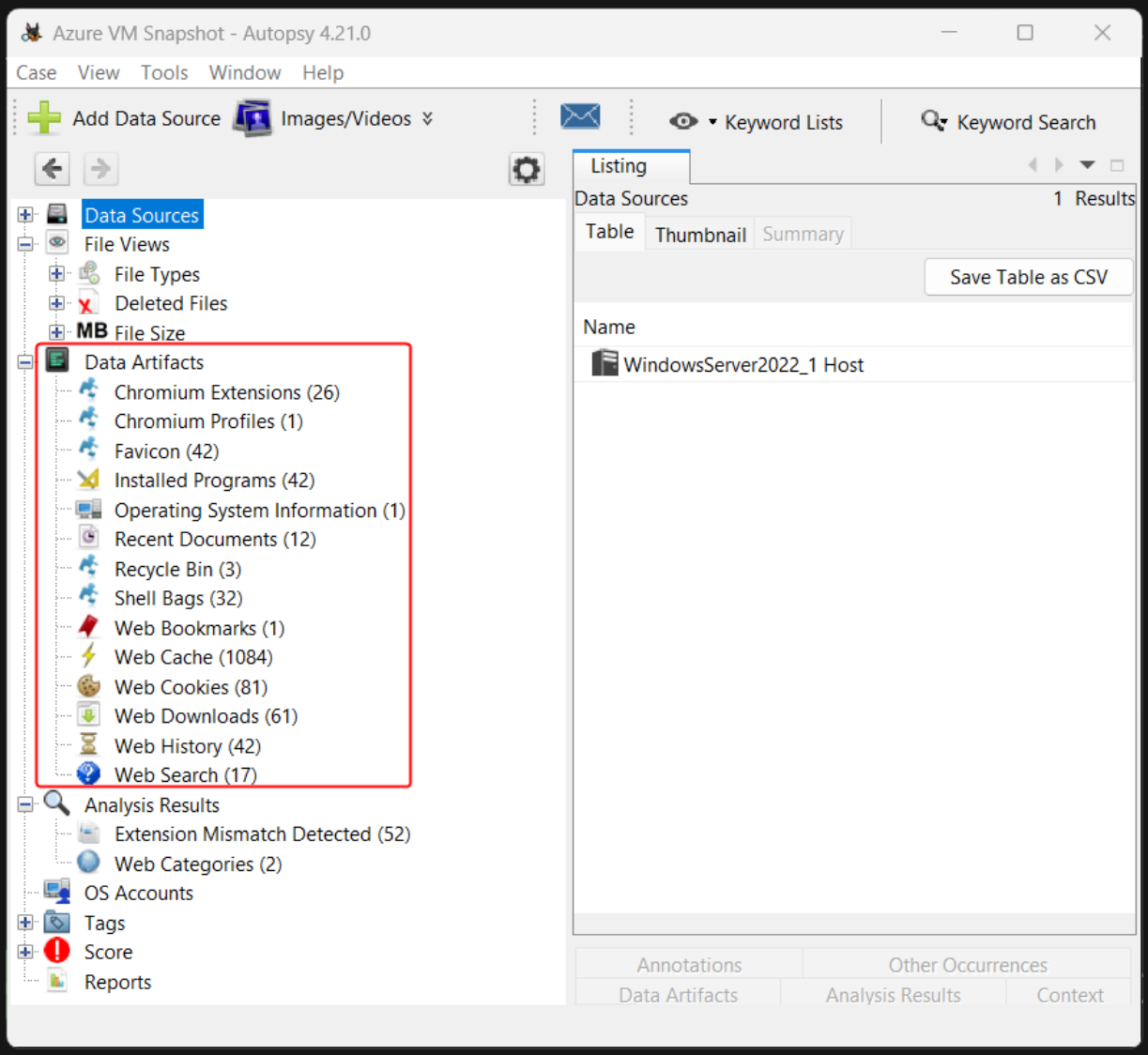


Figure 4.4 9 - The user interface of Autopsy showing the data artifacts.

The next chapter will conduct a forensic analysis of the findings found on this disk image.

5. Forensic Report of the Insider

The forensic report analyzes the digital evidence extracted from the logs of the storage account that was connected to Log Analytics. This analysis covers a detailed review of the log data captured, including details of storage operations, access attempts, modifications, and any suspicious activities detected within the storage environment.

The report also investigates the disk image from the virtual machine using Autopsy, to analyze the digital footprints left behind within the disk image.

5.1 Evidence from the Storage Account Logs

Chapter 4.2 demonstrated how to retrieve the failed operations of a storage account in Azure. By doing this, incident responders can identify potential indicators of compromise (IOC).

The following query is used to retrieve the failed operations:

```
StorageBlobLogs
| where TimeGenerated > ago(3d) and StatusText !contains "Success"
| project TimeGenerated, OperationName, Uri, RequesterUpn
```

Figure 5.1.1 is displaying one of the failed operations.

TimeGenerated [UTC]	2024-04-28T13:36:24.9181775Z
OperationName	PutBlob
Uri	https://conradiconsulting.blob.core.windows.net:443/documents/TradeSecrets.docx?_=1714311385185
RequesterUpn	Joe_Mole@jorn3006hotmail.onmicrosoft.com

Figure 5.1 1 – The first Put Blob operation.

OperationName is the name of the operation that was carried out.

The **Uri** column stands for Uniform resource identifier, and it shows the specific document that was accessed (*TradeSecrets.docx*).

The **RequesterUpn** column display the name of the requestor. (Baynash, *et al.* 2024)

UPN stands for User Principal Name; this is the identity of the user who made the operation. (NIST, 2020b)

So, this column is identifying Joe Mole as the user.

The operation **Put Blob** indicates that a blob (object) either has been uploaded to a storage account, or that an existing blob has been updated (edited). (McCullough, *et al.* 2023)

Conradi Consulting already have one existing document uploaded in their storage account named *TradeSecrets.docx*, which means that the user Joe Mole has tried to edit this document.

The operation name **Delete Blob** refers to a blob being deleted. *Figure 5.1.2* shows that Mr. Mole has tried to delete the document *TradeSecrets.docx.* (Sharma, *et al.* 2023)

TimeGenerated [UTC]	2024-04-28T13:37:13.9605629Z
OperationName	DeleteBlob
Uri	https://conradiconsulting.blob.core.windows.net:443/documents/TradeSecrets.docx?_=1714311434219
RequesterUpn	Joe_Mole@jorn3006hotmail.onmicrosoft.com

Figure 5.1 2 - The Delete Blob operation.

The last failed operation is shown in *figure 5.1.3*, and it is known that Conradi Consulting only have one document in their storage account. The **Put Blob** operation in this context means that Mr. Mole has tried to upload a software called *Mimikatz.exe.*

TimeGenerated [UTC]	2024-04-28T13:37:46.8365377Z
OperationName	PutBlob
Uri	https://conradiconsulting.blob.core.windows.net:443/documents/Mimikatz.exe?_=1714311466670
RequesterUpn	Joe_Mole@jorn3006hotmail.onmicrosoft.com

Figure 5.1 3 – The second Put Blob operation.

Mimikatz is a tool used for stealing the credentials of Windows users, like usernames and passwords. (MITRE ATT&CK, 2017)

5.2 The Evidence Found from the Disk Image

The data artifacts in Autopsy show the information from the extracted data source, and this is displayed in the left pane of *figure 5.2.1.* This report will look at some of the artifacts that has been extracted in Autopsy.

The Web Search

The web search is displaying all the search history entered by the user. This history includes details such as the search terms used, the search engine or platform utilized, and timestamps indicating when each search was performed.

Figure 5.2.1 shows that the user has searched the web on how to turn off windows security, methods on how to leak data, and anti-forensic tools, which are techniques used to prevent forensic analysis. (Perlman, 2022)

The web search also contains the search term keylogger. Keylogger is a program used to record the keystrokes that a user types in on a computer keyboard. (NIST, 2020a)

The timestamps that are highlighted in *figure 5.2.1* displays the time of the web search. The timestamp *2024-04-28* converges with the timestamps from the storage account logs, and indicates that these activities has been carried out on the same day.

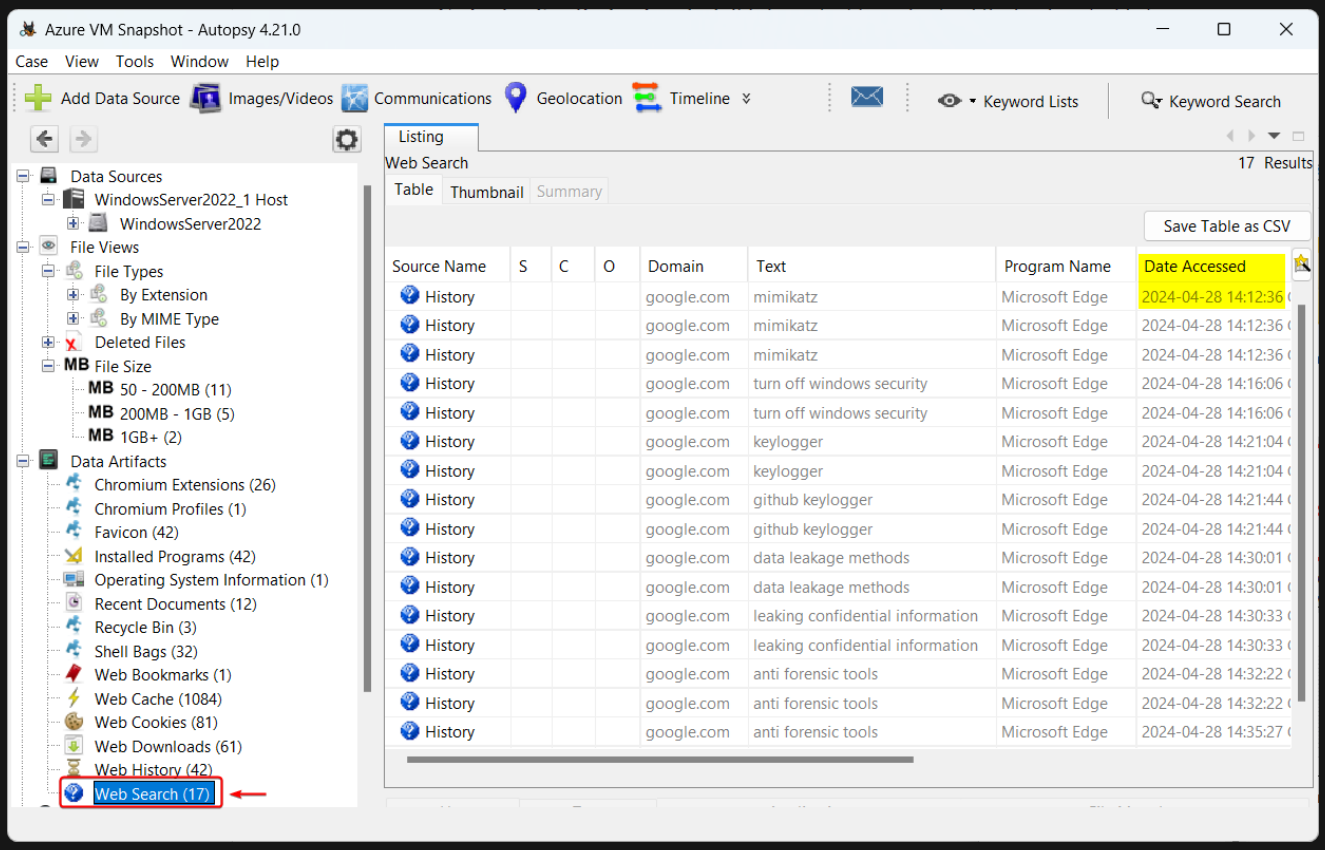


Figure 5.2 1 - The web search artifacts.

The Web History

The web browser history presents a detailed record of the different websites that the user has visited and interacted with while browsing the internet. It includes information such as the URLs of the visited websites.

By analyzing the web browser history, forensic investigators can gain valuable insights into the user's online behavior, interests, and activities.

One of these websites is Varonis.com, and *figure 5.2.2* reveals that the user has read up on Mimikatz.

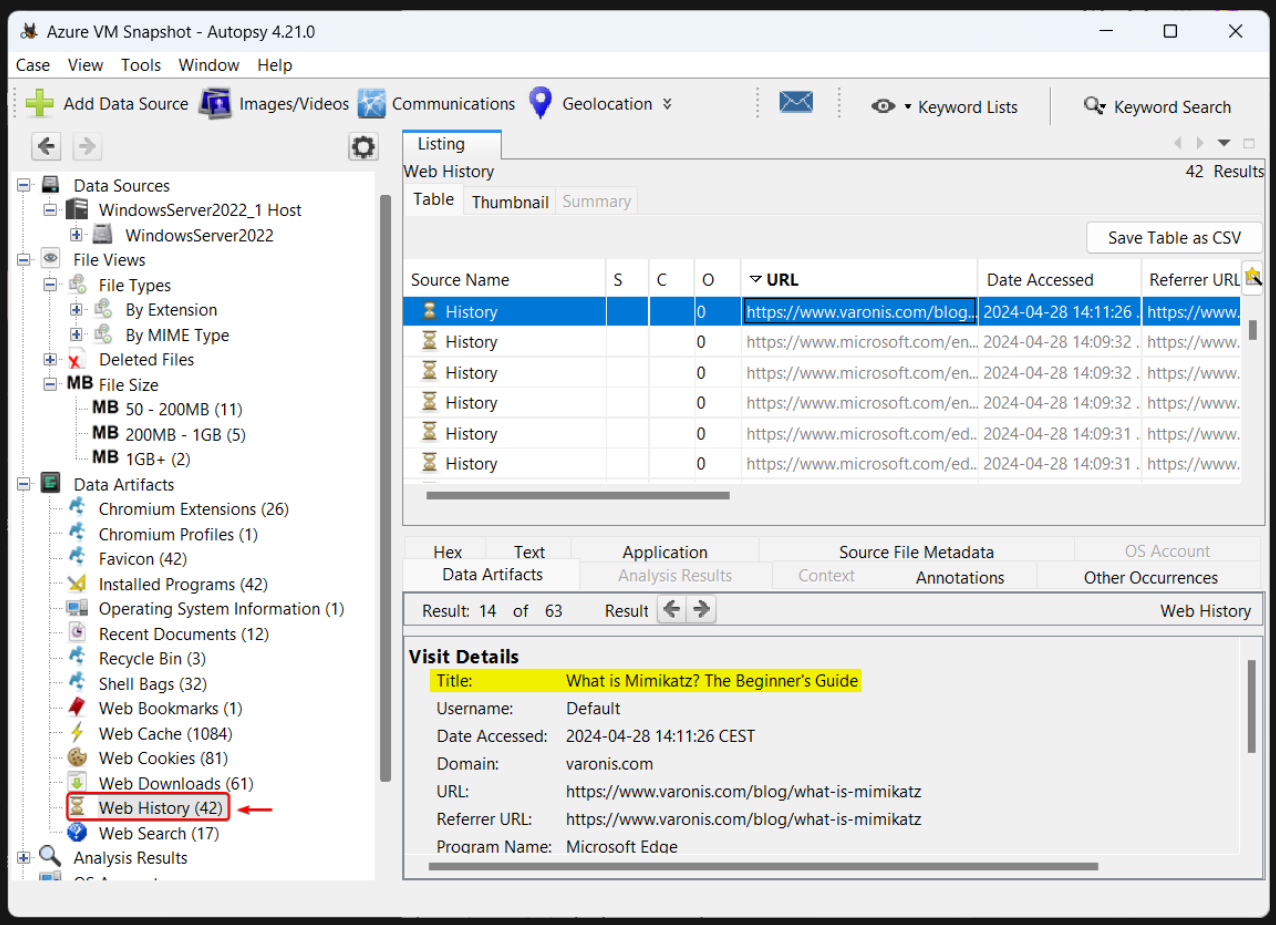


Figure 5.2 2 - The web history.

The Web Downloads

The web downloads tab is showing what the user has downloaded from the internet. The file path in *figure 5.2.3* reveals the username of the computer as Joe Mole. This is the same username that was exposed from the storage account logs.

The website downloads also show some of the software the user has downloaded, which is Mimikatz and Keylogger.

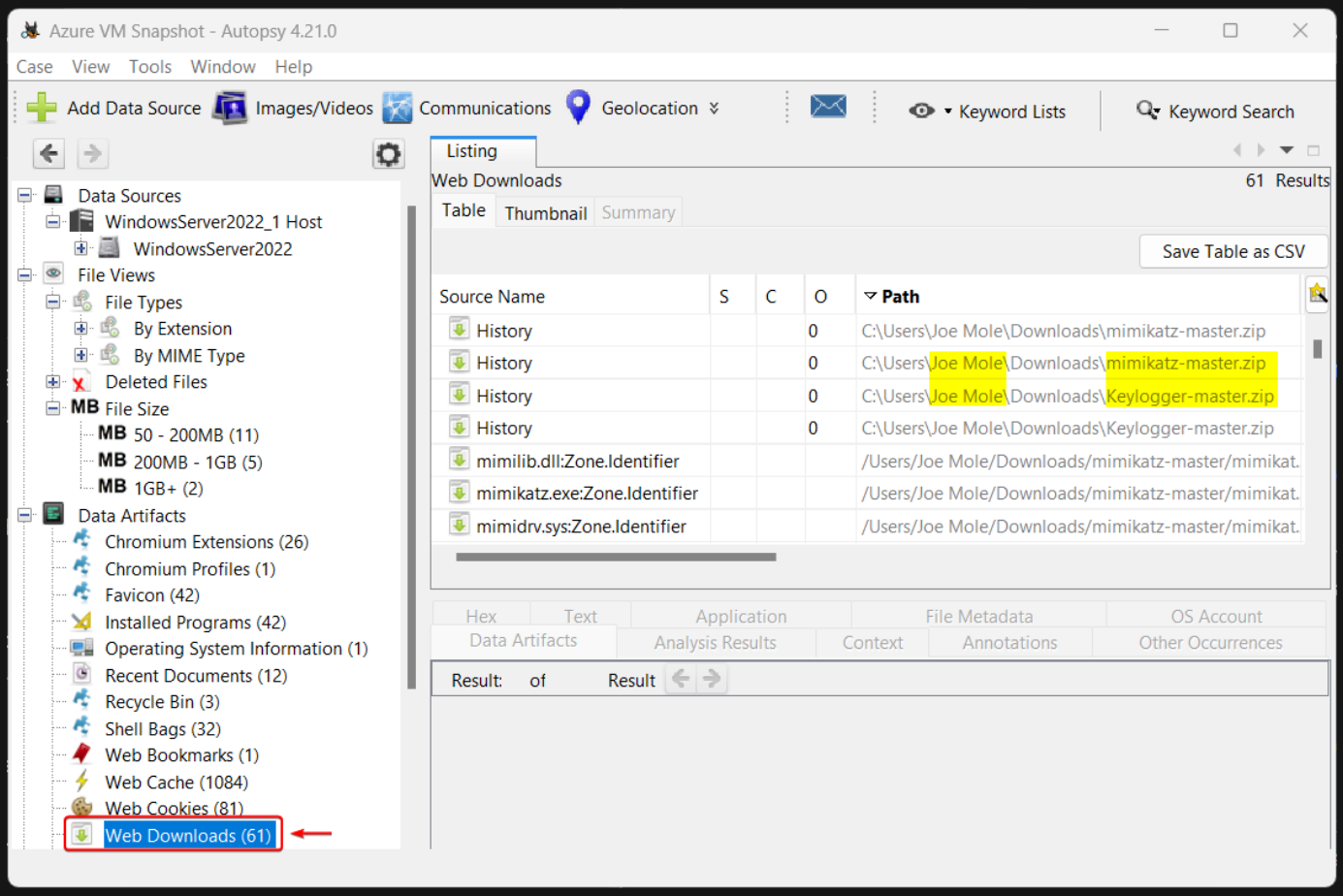


Figure 5.2 3 - The web downloads.

The Shell Bags

Shell Bags contain information about which folder a user has accessed and when this access took place. For example, if a company suspects an employee has deleted confidential information, Shell Bags can reveal if the computer of the employee was used to access this folder before this event occurred. (Lo, 2014)

Figure 5.2.4 indicates that the user has accessed the log directory on their computer.

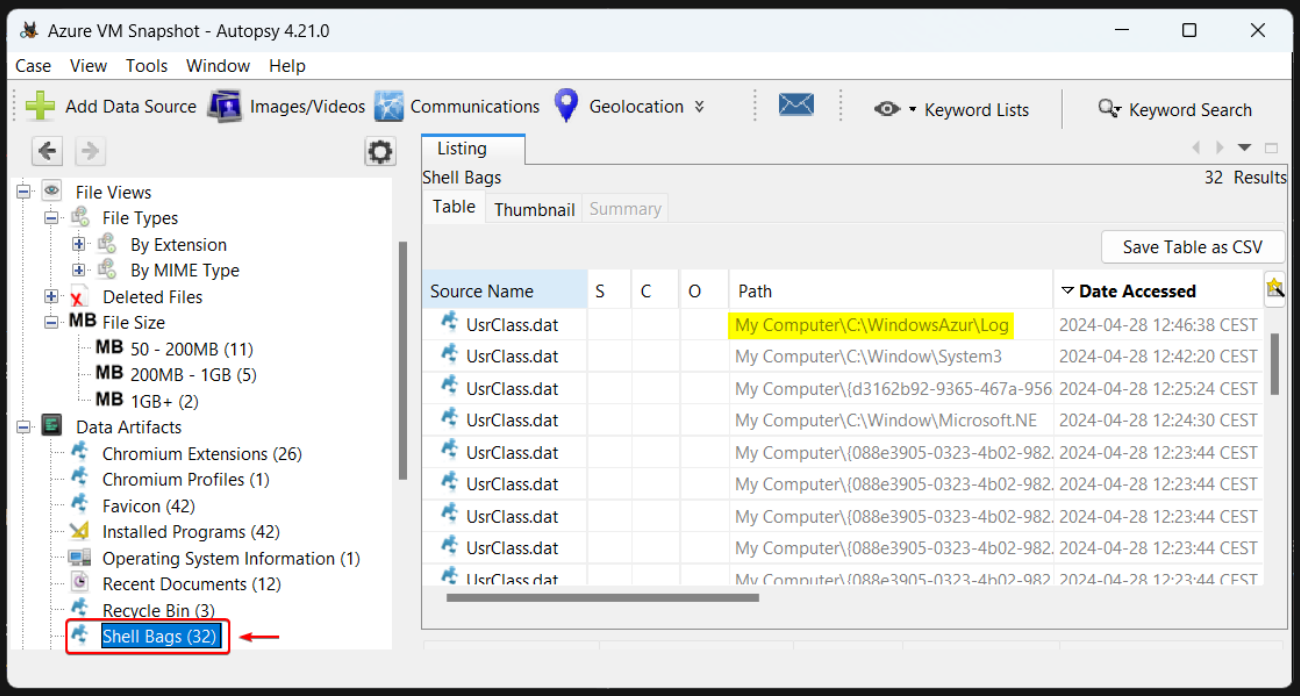


Figure 5.2 4 - The shell bags information.

The Recycle Bin

The deleted files from the user's computer are located in the recycle bin. The information in the recycle bin demonstrates that the user has deleted the keylogger that was previously discovered in the web downloads.

The highlighted text in *figure 5.2.5* shows that an *EVTX file* was deleted from the log directory. This file is a log of events that is created by Windows services, and it includes event data like operation types, process names, and timestamps, among other things. (FILExt.com, 2008)

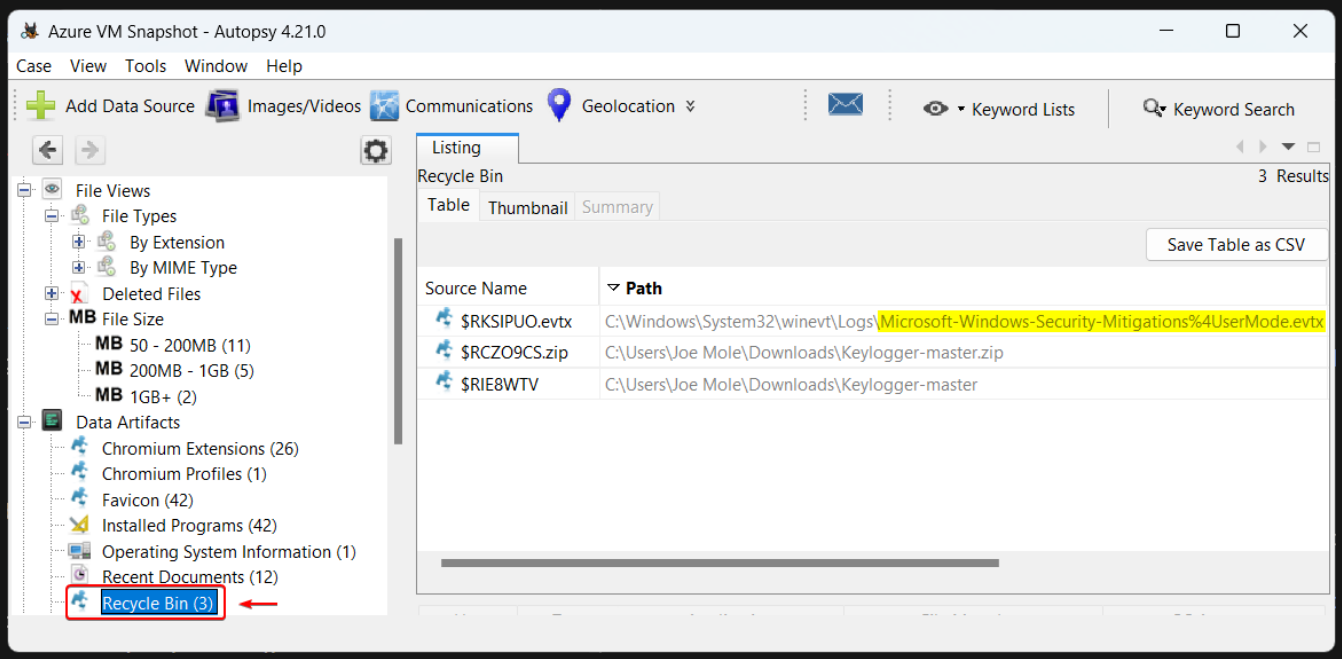


Figure 5.2 5 - The recycle bin of the computer.

The Recent Documents

The recent documents provide an overview of the various files and documents that the user has accessed and opened on their computer system. By examining these documents, investigators can gain insights into the types of files the users have been accessing.

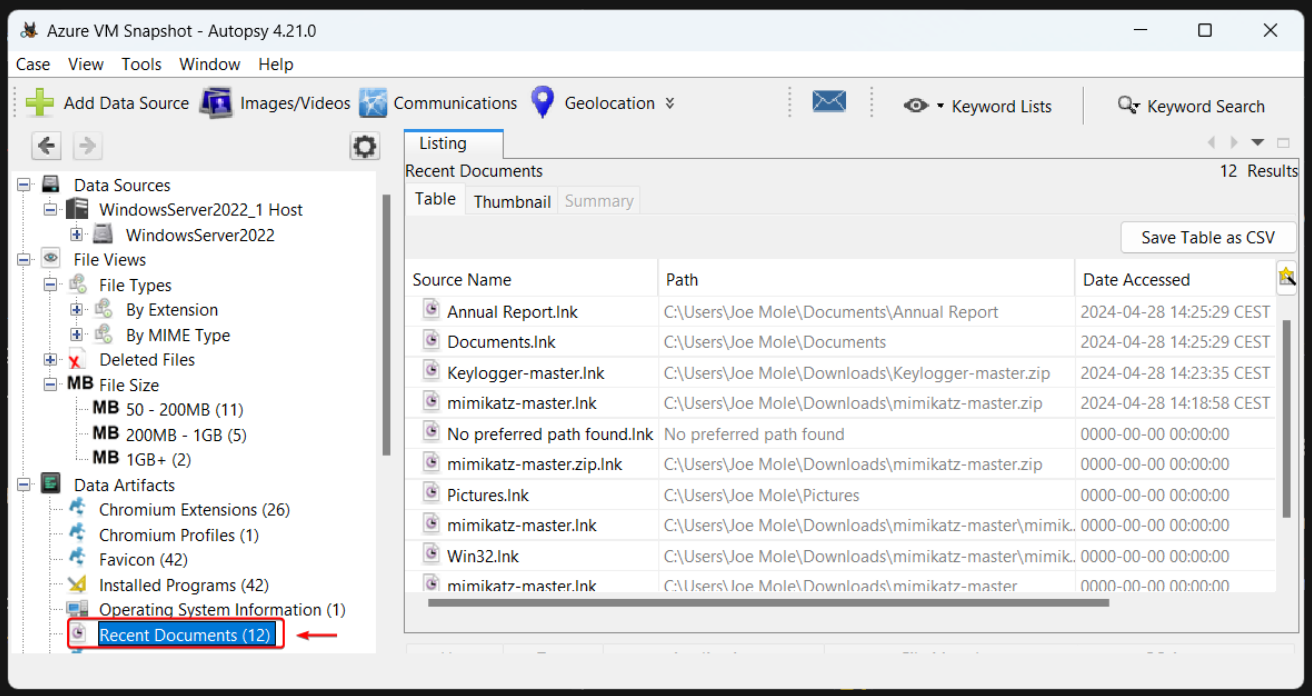


Figure 5.2 6 - The recent documents.

An investigator can start examining a file further by right clicking on the file and choose *View Associated File in Directory*, as seen in *figure 5.2.7.*

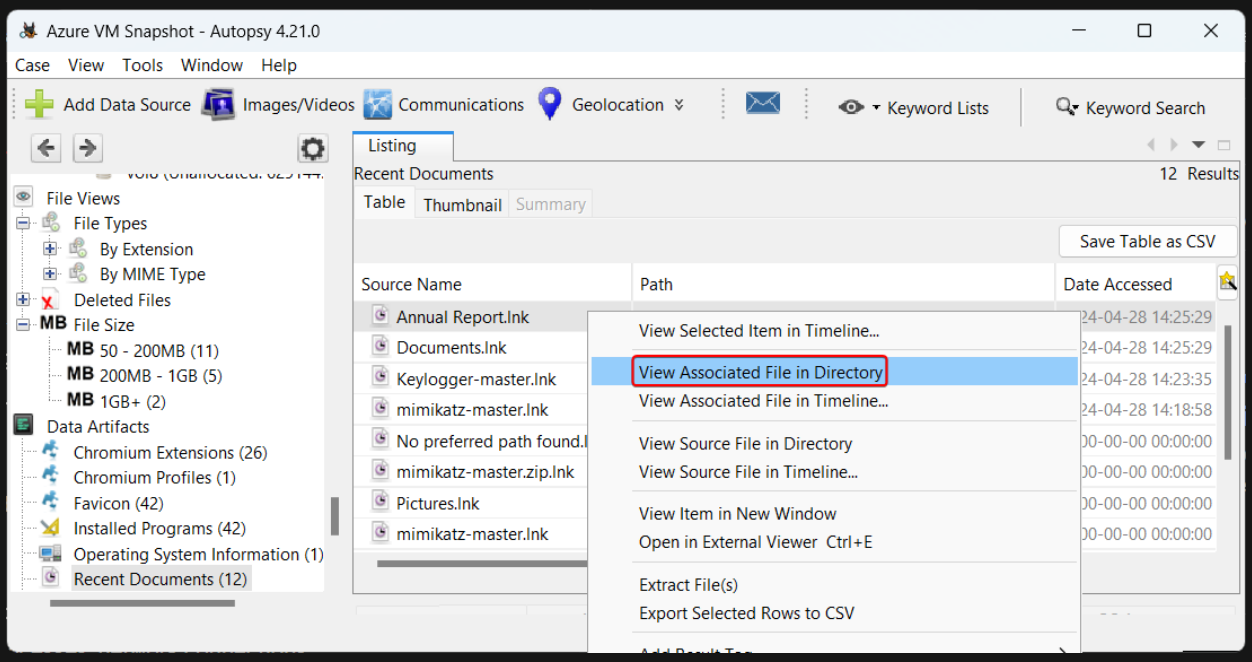


Figure 5.2 7 - Examining a specific file.

Proceed by clicking on the *Annual Report* folder.

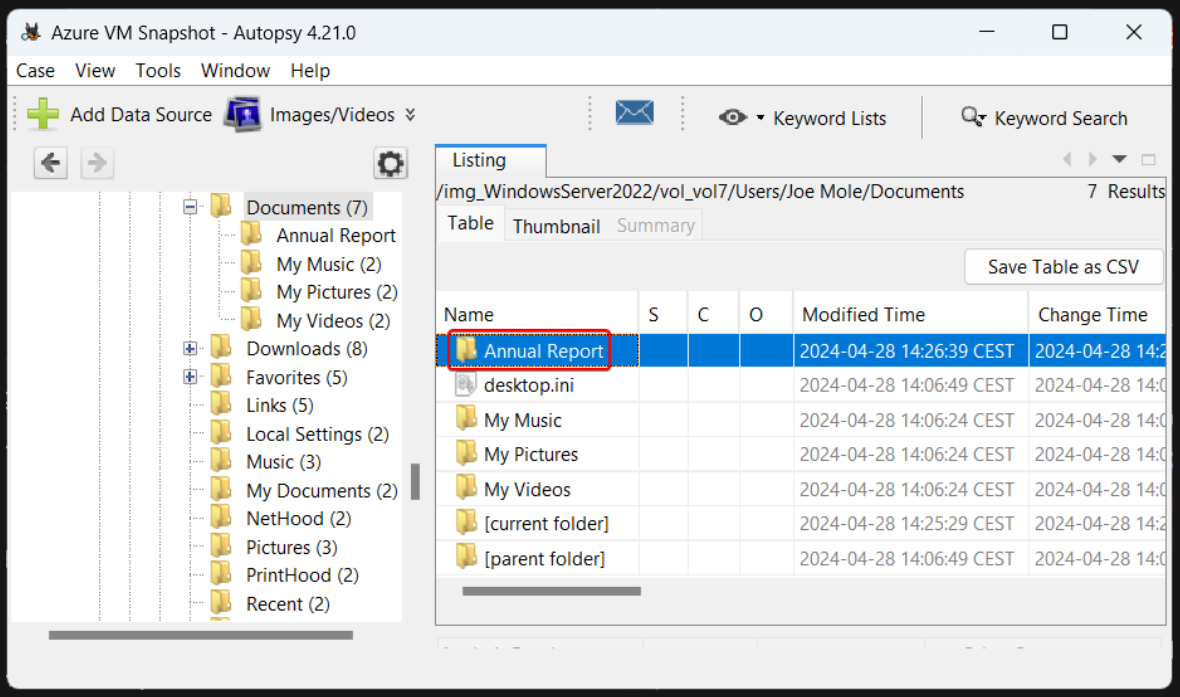


Figure 5.2 8 - Selecting the specific folder.

Figure 5.2.9 is displaying the file extension of the selected file. This file has a double extension in the filename: AnnualReport.docx.exe. Autopsy is showing the extension of the file as .exe (executable). Some people may think this is a document file by looking at the filename and the first extension, .docx.

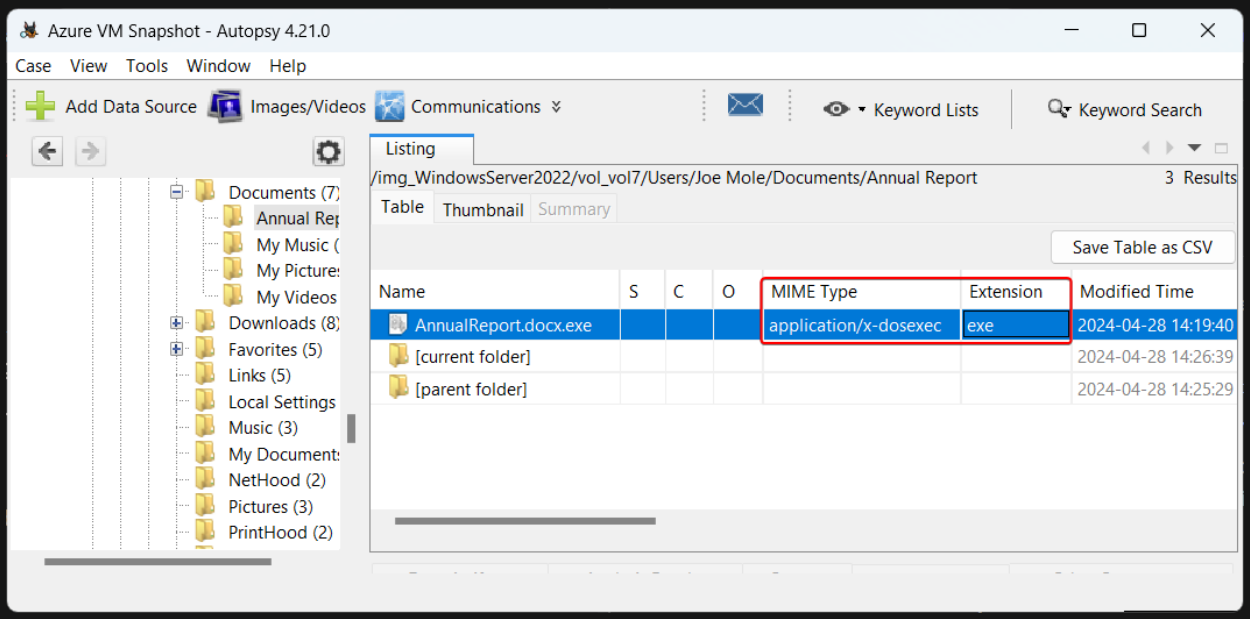


Figure 5.2 9 - The extension of the file.

The knowledge base MITRE ATT&CK helps cybersecurity practitioners, companies, and governments in identifying the strategies and tactics of cyber criminals.
(MITRE ATT&CK, 2020)

According to MITRE ATT&CK, this technique is known as masquerading. Attackers is using a double file extension to masquerade the real file type, trying to trick people into opening a malicious file. (MITRE ATT&CK, 2021)

In Autopsy, It is possible to extract the file for further investigation, but this can be risky if the file contains malware or other malicious content. It's generally safer to analyze suspicious files in a controlled environment, such as within a virtual machine.

Right click on the file and select **Extract File** to save it to the computer.

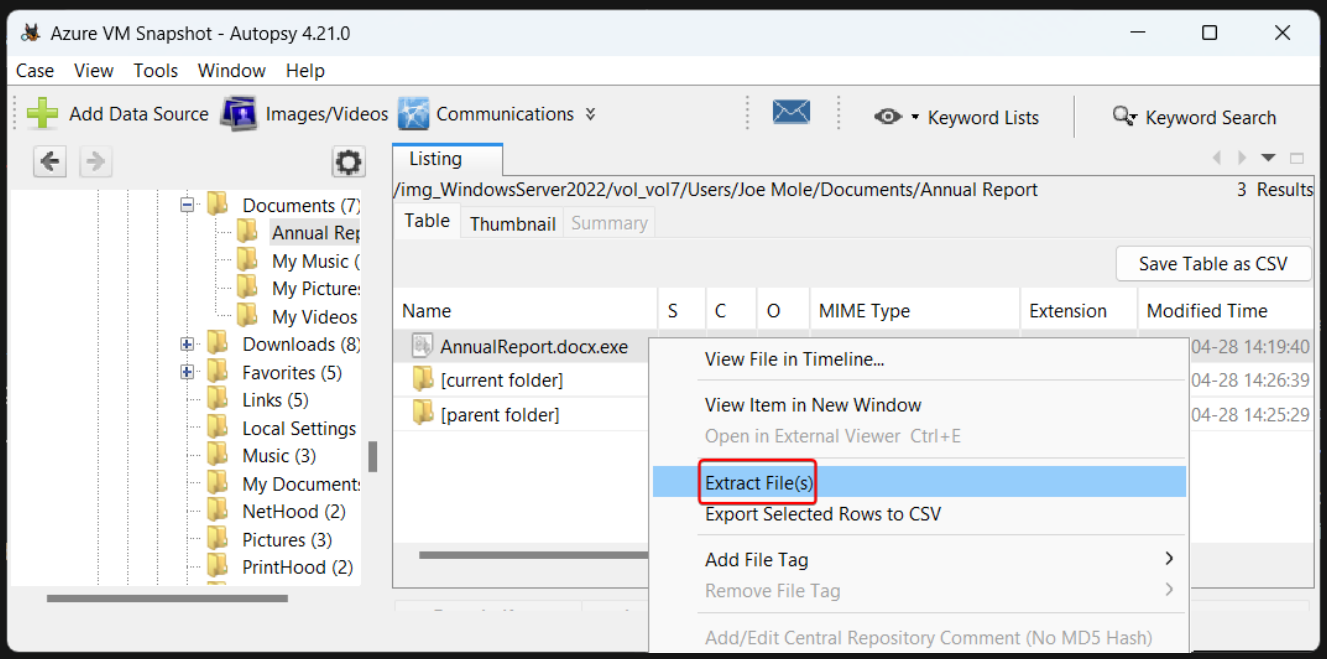


Figure 5.2 10 - Extracting the selected file.

Windows Security instantly quarantined the file following the file's extraction from Autopsy.

Figure 5.2.11 confirms that this is a hack tool by Windows Security. This is the same software (Mimikatz) that the user read about and downloaded. This finding demonstrates that the user has changed the filename and extension for the purpose of masquerading the original file type.

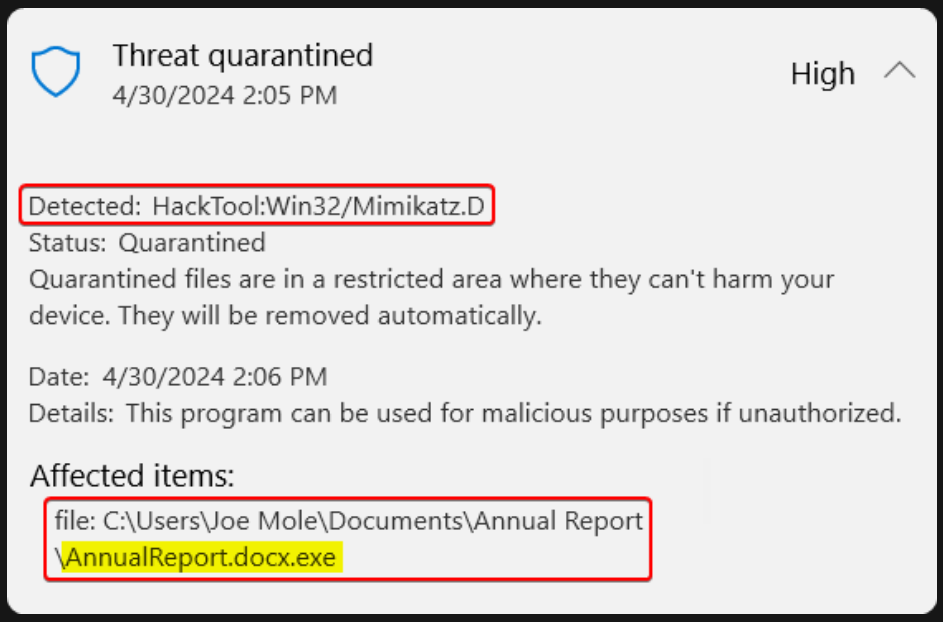


Figure 5.2 11 - The file quarantined by Windows Security.

6. Discussion and Analysis

6.1 Methods and Tools Used

By connecting an Azure storage account to Log Analytics, enables the monitoring of the storage account and the ability to track all storage operations occurring within the Azure environment.

A snapshot of the virtual machine was also taken. The disk image of the virtual machine was downloaded and extracted within Autopsy for further analysis.

This process allows investigators to thoroughly analyze the contents of the virtual machine's disk, including files and data artifacts, uncovering potential evidence of malicious activities.

6.2 Key Findings

Through detailed examination of the logs in Log Analytics, the insider was identified as Joe Mole. The logs revealed attempts to edit and delete a confidential document, along with efforts to upload password-stealing software (Mimikatz).

The digital evidence uncovered in Autopsy confirmed that the same individual downloaded the software Mimikatz and changed the filename and extension for the purpose of masquerading the filetype.

The timestamps from the storage account logs and the data artifacts discovered in Autopsy showed that the activity of the user was done on the same day.

Evidence from both sources clearly showed the same individual as the insider.

6.3 Lessons Learned

This investigation demonstrates the significant impact an insider threat can have on a company’s data security.

It also highlights how important it is to have reliable monitoring and forensic tools in detecting and addressing such threats. The effectiveness of Azure Log Analytics and Autopsy in this investigation demonstrates their value.

7. Conclusion

The primary objective of this project was to simulate an insider threat within a company and utilize forensic tools to identify the perpetrator.

7.1 Final Thoughts

In conclusion, this project successfully achieved its objectives, demonstrating the importance of being prepared and proactive in addressing insider threats. Continuous improvement of forensic capabilities is essential to maintaining robust security in any organization.

This ensures that the organization's digital assets are properly protected against potential threats and that any security incidents can be effectively investigated and addressed. Also, ongoing training and skill development enables forensic investigators to adapt to evolving cyber threats and take proactive steps to protect sensitive information and systems.

7.2 Recommendations

To prevent similar incidents in the future, it is recommended that Conradi Consulting should implement stricter access controls, such as limiting who can access sensitive data and systems. One of the methods to limit access to a storage account is to use the Role-Based Access Control (RBAC).

By using RBAC, Conradi Consulting can grant access by creating a role assignment for the users who need access to the resources. They can also revoke access by removing the users who should not have this access. This is done simply by removing the role assignment of the user. (Microsoft Learn, 2018)

Enhanced monitoring is also important, such as keeping a closer eye on user activities and network traffic to detect any suspicious behavior quickly. Azure Monitor is one of the tools which companies can use to detect suspicious behavior. By utilizing Azure Monitor, Conradi Consulting can effectively monitor their Azure resources and respond to system events. (Wren, *et al.* 2023)

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